national center for ontological research
ncor

University at Buffalo
The State University of New York

Future of Ontology Engineering

Dr. John Beverley

President, *National Center for Ontological Research, Inc.*
Assistant Professor, *University at Buffalo*
Associate Director, *Institute of AI and Data Science*
CEO, *Acacia Knowledge Systems Inc.*

Outline

- Research and Development
- The Structure of Scientific Revolutions
- A New Era

Outline

- Research and Development
- The Structure of Scientific Revolutions
- A New Era

Guiding Questions

How much alignment can you enforce across communities without undermining user goals?

What can be done to minimize downstream alignment efforts across diverging communities?

FY26 National Defense Authorization Act

SEC. 1504. DEPARTMENT OF DEFENSE DATA ONTOLOGY GOVERNANCE WORKING GROUP.

(a) ESTABLISHMENT.—

(1) IN GENERAL.—The Secretary of Defense shall establish a working group to develop and implement a common data ontology and governance structure across the Department of Defense.

(2) DESIGNATION.—The working group established under to paragraph (1) shall be known as the "Department of Defense Data Ontology Governance Working Group" (in this section the "Working Group").

(3) USE OF EXISTING STRUCTURES.—

(A) IN GENERAL.—Notwithstanding paragraph (1), the Secretary of Defense may designate an existing forum, council, or organizational body to serve as the Working Group if such entity satisfies the requirements of sub-sections (b) and (c).

(B) RULE OF CONSTRUCTION.—For the purposes of this section, a forum, council, or organizational body designated under subparagraph (A) is deemed to be a working group established by the Secretary of Defense under paragraph (1).

(b) PURPOSE.—The purpose of the Working Group is to inform and to progress the Department of Defense's foundational data ontology work by developing and implementing domain-specific data ontologies and governance structures across the Department of Defense to expand data interoperability, enhance information

Semantic Layer vs Operational Layer

Meaning is governed here; behavior is implemented there.

Semantic Layer

- Authoritative meaning
- Classes, relations, and definitions
- Identifiers, axioms, and mappings
- Validation and provenance patterns
- Governance, versioning, and release control

Defines what terms mean and how they relate.

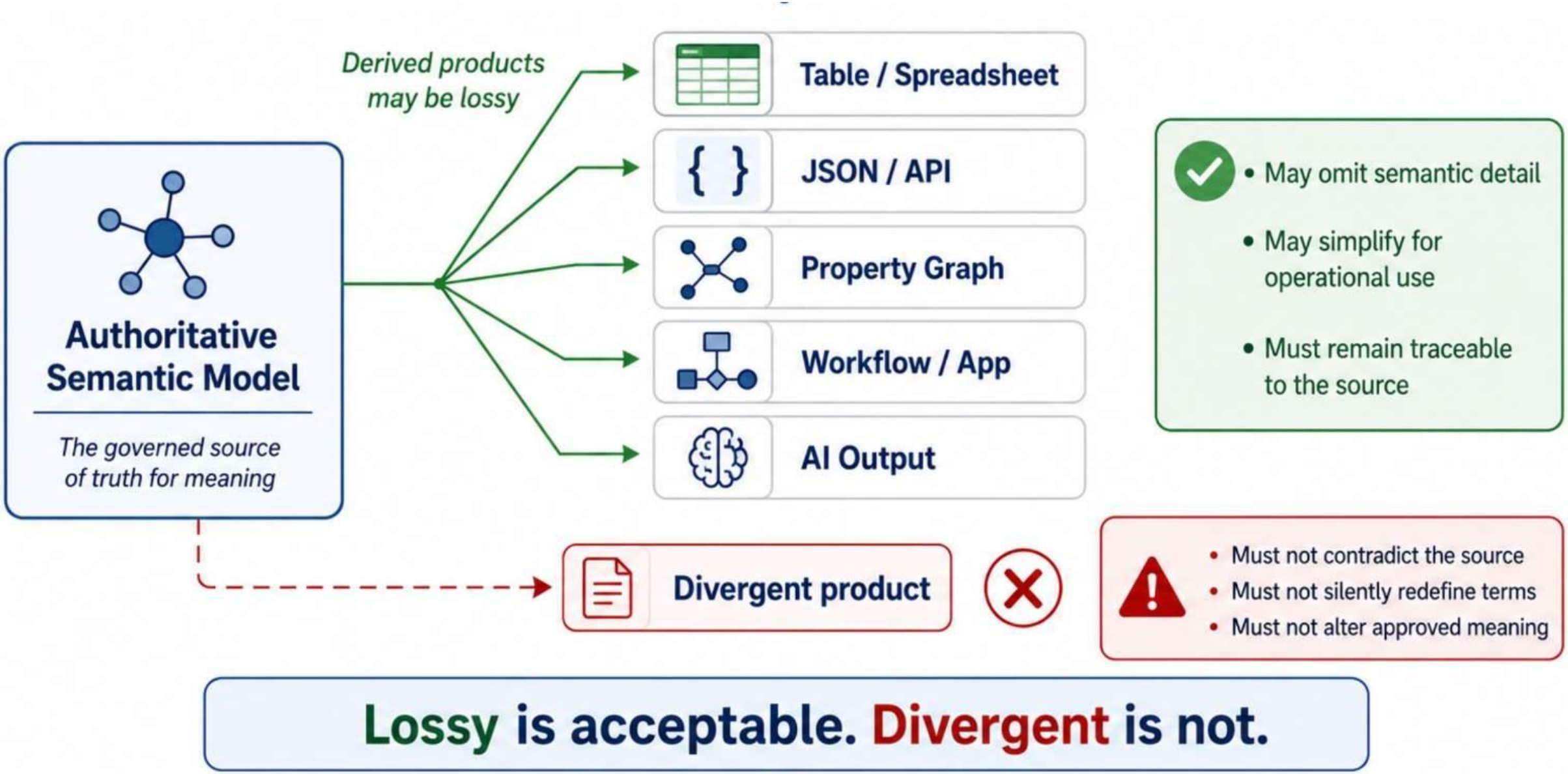
Traceable, governed derivation

Operational Layer

- Applications and workflows
- Databases, APIs, and user interfaces
- Analytics, dashboards, and search
- AI tools and operational automation
- Performance, scale, and user experience

Implements mission behavior using derived representations.

Meaning → **Execution**


Derived products may be lossy
Authoritative Semantic Model
The governed source of truth for meaning
Table / Spreadsheet
JSON / API
Property Graph
Workflow / App
AI Output
May omit semantic detail
May simplify for operational use
Must remain traceable to the source
Divergent product
Must not contradict the source
Must not silently redefine terms
Must not alter approved meaning
Lossy is acceptable. Divergent is not.

Guiding Questions

How much alignment can you enforce across communities without undermining user goals?

What can be done to minimize downstream alignment efforts across diverging communities?

History of Mapping: 90s – 00s

- Emerged from the knowledge representation and AI communities, largely driven by the need to integrate heterogeneous knowledge bases

- Strategies were manual: experts compared class hierarchies, identified overlaps, and wrote crosswalks or bridging axioms between ontologies

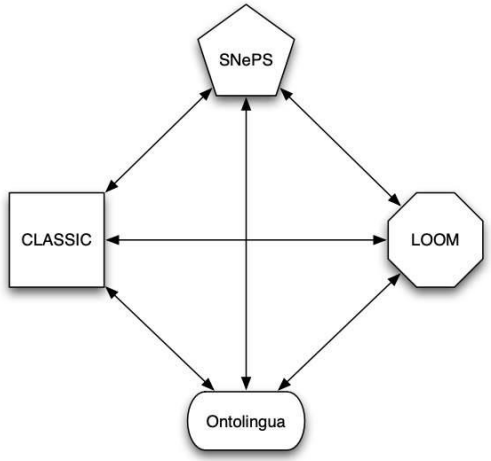
- Early efforts (e.g., UMLS Metathesaurus, Gene Ontology cross-links) focused on lexical matching (shared labels, synonyms) and human curation

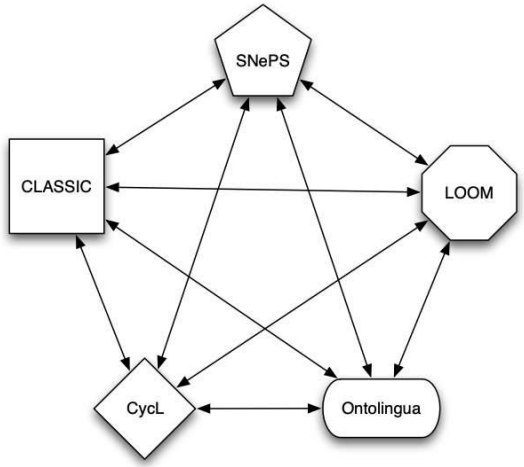
Minimizing Mappings

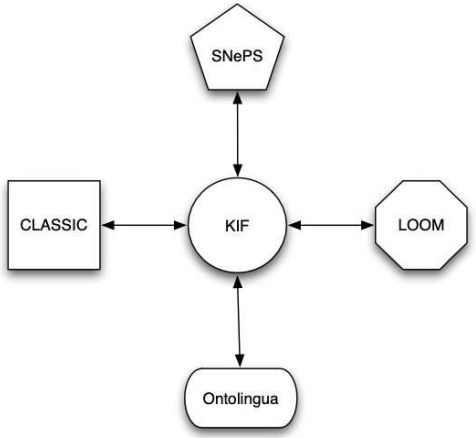
- Connecting disparate datasets requires two-way mappings:
  - 2 datasets – 2 mappings
  - 3 datasets – 6 mappings
  - 4 datasets – 12 mappings

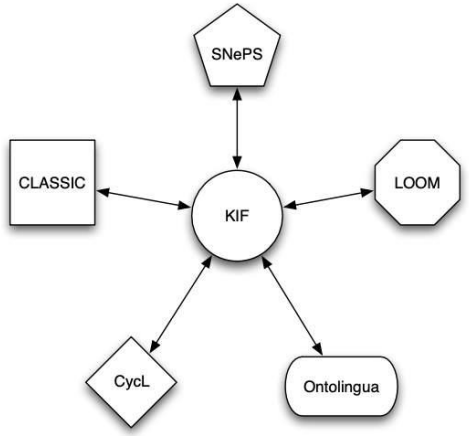
    ....

- Ontologies with common semantics help minimize the number of needed mappings for interoperability









History of Mapping: 05s

- Semantic Web stack matured with tools like PROMPT (Protégé plugin) and FOAM

- Strategies emphasized:
  - Lexical similarity (string, edit distance, WordNet expansion)
  - Structural similarity (aligning hierarchies, subsumption patterns)
  - Instance-based similarity (when data annotations were available)

- This period introduced the formalization of benchmarks and comparing system performance

History of Mapping: 2005 - 2015

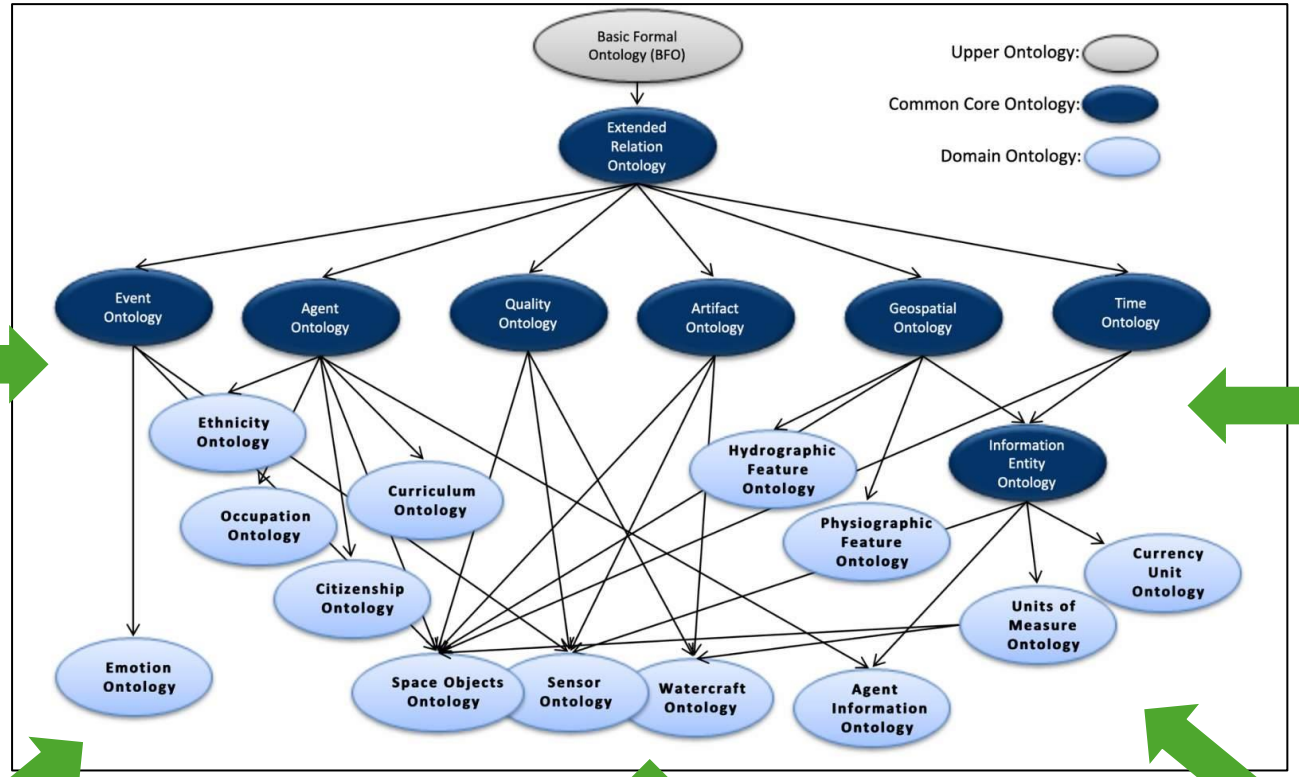
- Upper level ontologies such as **Basic Formal Ontology (BFO)**, were increasingly used as **anchors for alignment**

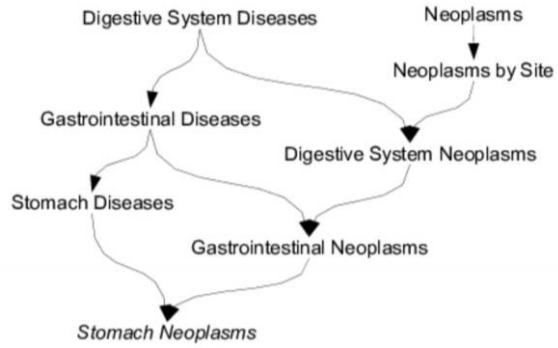
- Committing domain ontologies to a common top-level framework, reducing the number of arbitrary cross-maps

- For example, the **OBO Foundry** adopted BFO as a unifying basis, which meant mappings were often recast as **subclass axioms under BFO categories**

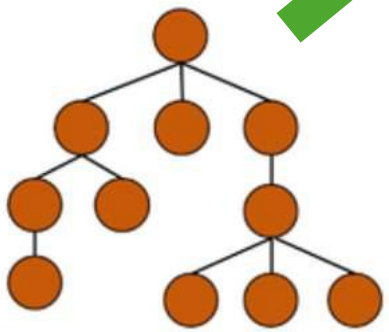


Bag of Words

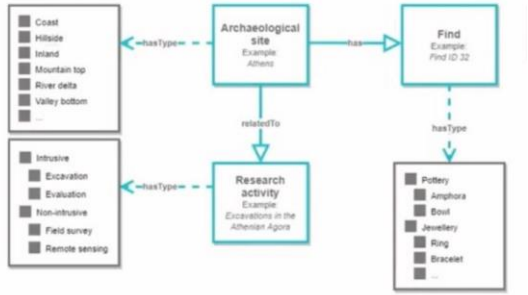




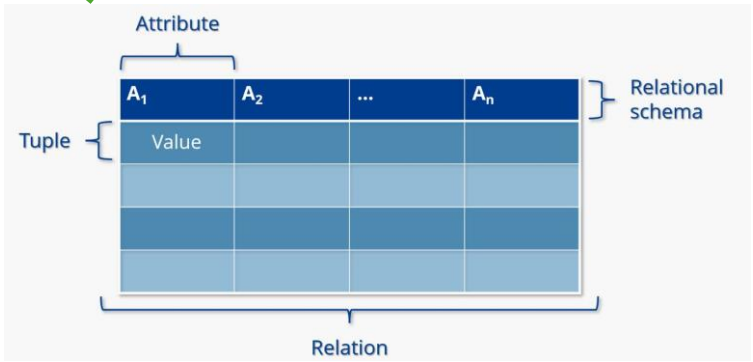
Thesaurus



Ontology



Controlled Vocabulary



Relational Database

gistBFO: An Open-Source, BFO-Compatible Version of gist

Dylan ABNEY [a,1], Katherine STUDZINSKI [a], Giacomo DE COLLE [b,c], Finn WILSON [b,c], Federico DONATO [b,c], and John BEVERLEY [b,c]
[a] *Semantic Arts, Inc.*
[b] *University at Buffalo*
[c] *National Center for Ontological Research*
ORCiD ID: Dylan Abney https://orcid.org/0009-0005-4832-2900, Katherine Studzinski https://orcid.org/0009-0001-3933-0643, Giacomo De Colle https://orcid.org/0000-0002-3600-6506, Finn Wilson https://orcid.org/0009-0002-7282-0836, Federico Donato https://orcid.org/0009-0001-6600-240X, John Beverley https://orcid.org/0000-0002-1118-1738

Abstract. gist is an open-source, business-focused ontology actively developed by Semantic Arts. Its lightweight design and use of everyday terminology has made it a useful tool for kickstarting domain ontology development in a range of areas including finance, government, and pharmaceuticals. The Basic Formal Ontology (BFO) is an ISO/IEC standard upper ontology that has similarly found practical application across a variety of domains, especially biomedicine and defense. Given its demonstrated utility, BFO was recently adopted as a baseline standard in the U.S. Department of Defense and Intelligence Community.

Because BFO sits at a higher level of abstraction than gist, we see an opportunity to align gist with BFO and get the benefits of both: one can kickstart domain ontology development with gist, all the while maintaining an alignment with the BFO standard. This paper presents such an alignment, which consists primarily of subclass relations from gist classes to BFO classes and includes some subproperty axioms. The union of gist, BFO, and this alignment is what we call "gistBFO." The upshot is that one can model instance data using gist and then instances of gist classes can be mapped to BFO. This not only achieves compliance with the BFO standard; it also enables interoperability with other domains already modeled using BFO. We describe a methodology for aligning gist and BFO, provide rationale for decisions we made about mappings, and detail a vision for future development.

Keywords. Ontology, upper ontology, ontology alignment, gist, BFO

Article | Open access | Published: 17 February 2025

A semantic approach to mapping the Provenance Ontology to Basic Formal Ontology

Tim Prudhomme ✉, Giacomo De Colle, Austin Liebers, Alec Sculley, Peihong "Karl" Xie, Sydney Cohen & John Beverley

Scientific Data **12**, Article number: 282 (2025) | Cite this article

5472 Accesses | **4** Citations | **8** Altmetric | Metrics

Abstract

The Provenance Ontology (PROV-O) is a World Wide Web Consortium (W3C) recommended ontology used to structure data about provenance across a wide variety of domains. Basic Formal Ontology (BFO) is a top-level ontology ISO/IEC standard used to structure a wide variety of ontologies, such as the OBO Foundry ontologies and the Common Core Ontologies (CCO). To enhance interoperability between these two ontologies, their extensions, and data organized by them, a mapping methodology and set of alignments are presented according to specific criteria which prioritize semantic and logical principles. The ontology alignments are evaluated by checking their logical consistency with canonical examples of PROV-O instances and querying terms that do not satisfy the alignment criteria as formalized in SPARQL. A variety of semantic web technologies are used in support of FAIR (Findable, Accessible, Interoperable, Reusable) principles.

FOWG Subgroup Report

IES-BFO Cross-Ontology Working Group

FOUST

Authors	Title
Accepted Papers	
Laure Vieu and Adrien Barton	Order in the Mereology of Slots
Brandon Bennett	Building Up: foundations and material for definitional ontology construction
Fumiaki Toyoshima and Ludger Jansen	Malfunctioning artifacts: A step towards a realizable-centered unifying account
Fumiaki Toyoshima and Satoru Niki	Subsumption in the Mirror of Ontological and Logical Choices
Ian Bailey, John Beverley, Helene Blackmore, Andreas Cola, Paul Cripps, Giacomo de Colle, Federico Donato, Amanda Hicks, David Limbaugh, Elena Milivinti, Chris Patridge, Rebecca Rafferty and Barry Smith	Comparing Information Exchange Standard and Basic Formal Ontology Design Patterns
Michel Dumontier, Remzi Çelebi, Komal Gilani, Isabelle de Zegher, Katerina Serafimova, Catalina Martínez Costa and Stefan Schulz	SULO — a simplified upper-level ontology
Lucas V. Vieira, Cauã R. Antunes, Mara Abel, Fabrício H. Rodigues and Lisa Stright	Towards a reference ontology of the spatial location of physical objects

https://www.dmi.unict.it/fois2025/?page_id=867

History of Mapping: 2005 - 2015

- Instead of one-off mappings, ontologists began creating **import modules** (via MIREOT, OntoFox, ROBOT extract) to bring terms across ontologies

- **Bridging ontologies** were created e.g., for anatomy across species, or disease–phenotype mappings

- In biomedicine, major mapping efforts included:
  - **Ontology of Biomedical Investigations (OBI)** importing across OBO
  - SNOMED CT, LOINC, and other clinical terminologies to research ontologies

History of Mapping: 2015 - Present

- With the explosion of biomedical data and knowledge graphs, mapping began leveraging **NLP**, **embeddings**, and **ML-based similarity**

- Strategies include:
  - **Vector-space alignment**: word embeddings (e.g., BioBERT, SciBERT) to propose mappings
  - **Graph embeddings**: aligning ontologies based on structural patterns
  - **Hybrid pipelines**: combining lexical, structural, and ML similarity scores, filtered by logical consistency checks

- Example: **BioPortal mappings**

Ambiguity

- Despite a cottage industry of research on this topic, the field is underdeveloped with few obvious successes

- This seems partly to do with deep ambiguity at the heart of the research program

- In short: “mapping” can mean many things to many people

SSSOM

- Simple Standard for Sharing Ontological Mappings is a lightweight, tabular format (usually TSV/CSV) for representing and exchanging ontology mappings

- Each row records a mapping between two entities, with fields for IDs, labels, mapping predicate (e.g., skos:exactMatch), and provenance (creator, method, confidence)

- Considered a “mapping”

Formal Equivalence

- Such mappings bury semantics in strings such as “narrow match” making them opaque to machines

- When I think of mappings being useful, it is insofar as they can be used in specific applications or as a foundation for code

- Which is to say, the semantics must be spelled out carefully

Ontology Mapping

- Ontology mappings **matter most** where correctness, invariance, and interoperability are not optional

- Ontology mapping provides a **guarantee** that systems using the mapping can meaningfully reason with, transform, or integrate each other's data

- It's not a thesaurus; it's a **bridge between ontological commitments**

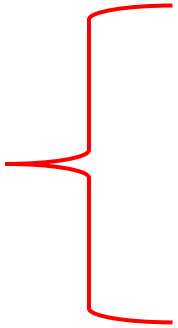
Ontology Mapping in Application

- Best understood in terms of use in an application
- Ontology mappings should not be evaluated using the same standards we use when evaluating ontologies
- For example, an IES-BFO mapping need not itself be a well-formed ontology to reflect common ground between these ontologies
- To be useful in applications, mappings **should be rigorous**

Ontology Map

- An **ontology map** is a statement <s, p, o> such that:
  - ‘s’ is a subject term representing a class or object property in a ontology,
  - ‘o’ is an object term representing a class or object property in some other ontology,
  - ‘p’ is a predicate that specifies how s and o relate

Basic Formal Ontology (BFO)



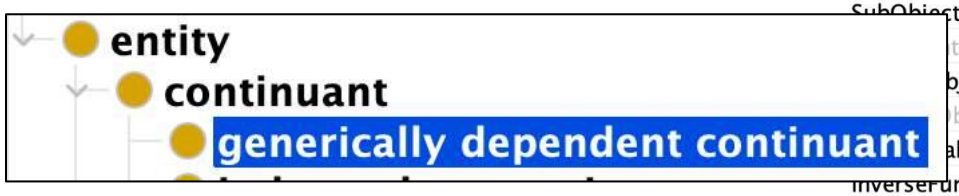
Metrics	
Axiom	602
Logical axiom count	192
Declaration axioms count	87
Class count	36
Object property count	40
Data property count	0
Individual count	0
Annotation Property count	13

Class axioms	
SubClassOf	66
EquivalentClasses	0
DisjointClasses	10
GCI count	0
Hidden GCI Count	0

Object property axioms	
SubObjectPropertyOf	6
EquivalentObjectProperties	0
InverseObjectProperties	17
DisjointObjectProperties	0
FunctionalObjectProperty	6
InverseFunctionalObjectProperty	1
TransitiveObjectProperty	6
SymmetricObjectProperty	0
AsymmetricObjectProperty	0
ReflexiveObjectProperty	0
IrrefexiveObjectProperty	0
ObjectPropertyDomain	40
ObjectPropertyRange	40
SubPropertyChainOf	0

Basic Formal Ontology (BFO)

Metrics	
Axiom	602
Logical axiom count	192
Declaration axioms count	87
Class count	36
Object property count	40
Data property count	0
Individual count	0
Annotation Property count	13
Class axioms	
SubClassOf	66
EquivalentClasses	0
DisjointClasses	10
GCI count	0
Hidden GCI Count	0
Object property axioms	
SubObjectPropertyOf	6
tObjectProperties	0
bjectProperties	17
bjectProperties	0
alObjectProperty	6
InverseFunctionalObjectProperty	1
TransitiveObjectProperty	6
	0
	0
	0
	0
	40
ObjectPropertyRange	40
SubPropertyChainOf	0



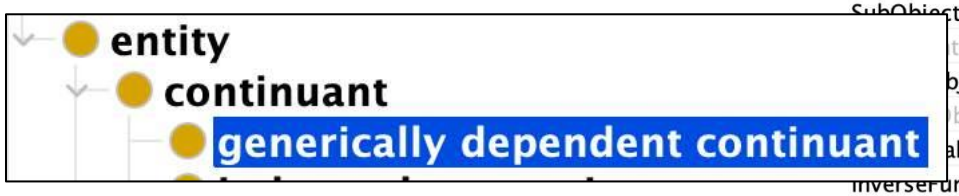


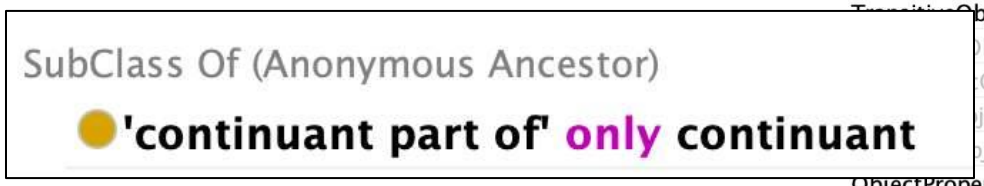
Basic Formal Ontology (BFO)

Metrics	
Axiom	602
Logical axiom count	192
Declaration axioms count	87
Class count	36
Object property count	40
Data property count	0
Individual count	0
Annotation Property count	13

Class axioms	
SubClassOf	66
EquivalentClasses	0
DisjointClasses	10
GCI count	0
Hidden GCI Count	0

Object property axioms	
SubObjectPropertyOf	6
tObjectProperties	0
bjectProperties	17
bjectProperties	0
alObjectProperty	6
InverseFunctionalObjectProperty	1
TransitiveObjectProperty	6
bjectProperty	0
ObjectProperty	0
jectProperty	0
bjectProperty	0
ObjectPropertyDomain	40
ObjectPropertyRange	40
SubPropertyChainOf	0





Basic Formal Ontology (BFO)

has occurrent part

Inverse Of
- 'occurrent part of'

Domains (intersection)
- occurrent

Ranges (intersection)
- occurrent

Metrics	
Axiom	602
Logical axiom count	192
Declaration axioms count	87
Class count	36
Object property count	40
Data property count	0
Individual count	0
Annotation Property count	13

Class axioms	
SubClassOf	66
EquivalentClasses	0
DisjointClasses	10
GCI count	0
Hidden GCI Count	0

Object property axioms	
SubObjectPropertyOf	6
EquivalentObjectProperties	0
InverseObjectProperties	17
DisjointObjectProperties	0
FunctionalObjectProperty	6
InverseFunctionalObjectProperty	1
TransitiveObjectProperty	6
SymmetricObjectProperty	0
AsymmetricObjectProperty	0
ReflexiveObjectProperty	0
IrrefexiveObjectProperty	0
ObjectPropertyDomain	40
ObjectPropertyRange	40
SubPropertyChainOf	0

- [ ] Functional
- [ ] Inverse functional
- [x] Transitive
- [ ] Symmetric
- [ ] Asymmetric
- [ ] Reflexive
- [ ] Irreflexive

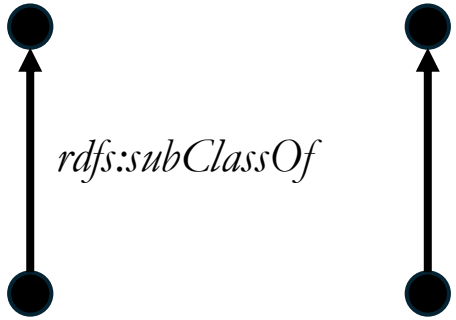
Axiom Patterns in BFO

- rdfs:subClassOf enforcing monohierarchy; disjointness enforced between most sibling classes

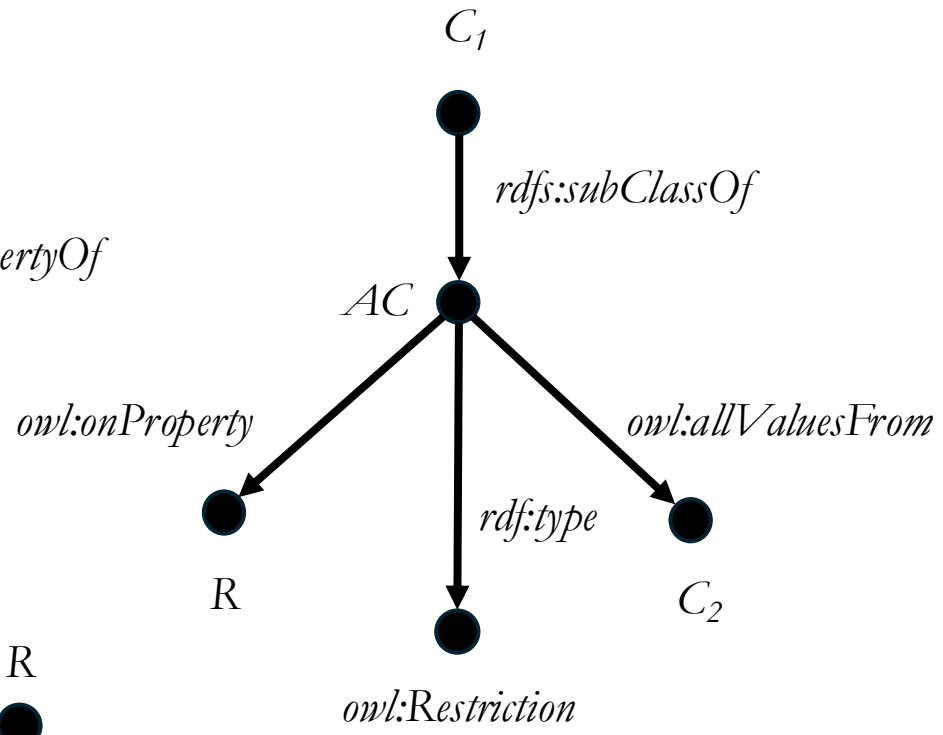
- No existential restrictions; only universal restrictions, e.g. **process occurrent part of only process**; several contain disjunctive ranges

- Nearly all object properties have domain/range, inverses; handful have characteristics, e.g. functional, transitive, etc.

BFO Axiom Patterns

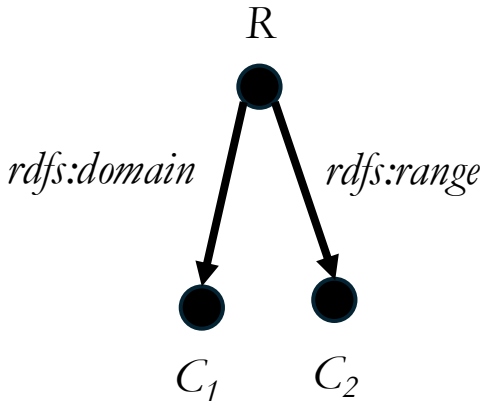


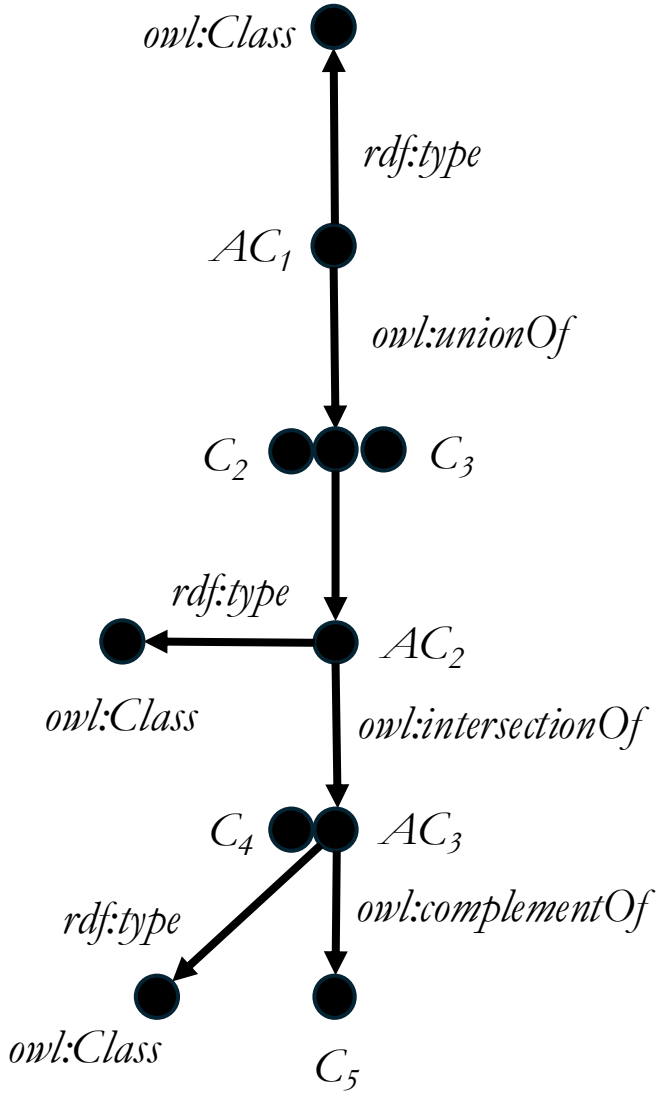






no rdfs:subClassOf
in either direction

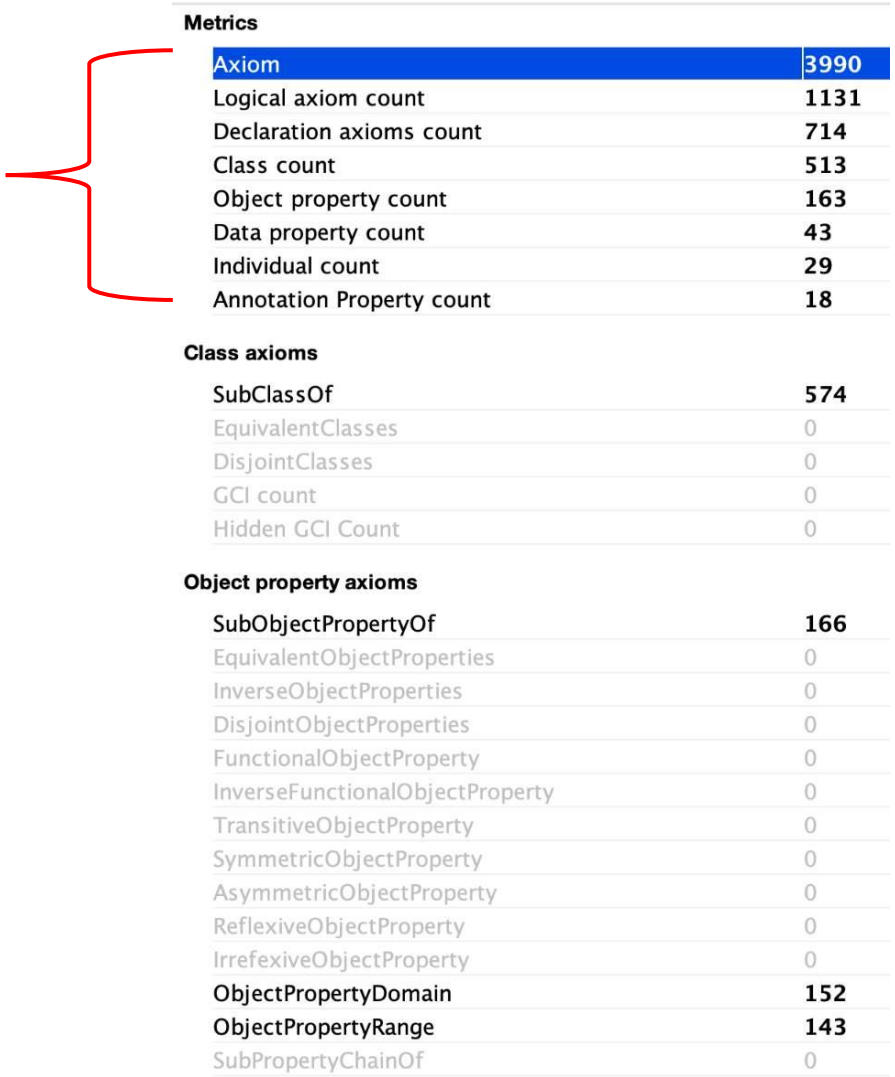




Information Exchange Standard (IES)

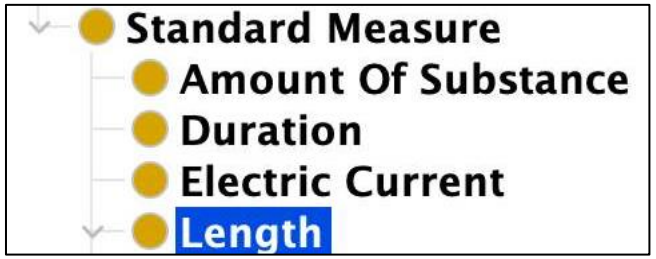
Metrics	
Axiom	3990
Logical axiom count	1131
Declaration axioms count	714
Class count	513
Object property count	163
Data property count	43
Individual count	29
Annotation Property count	18
Class axioms	
SubClassOf	574
EquivalentClasses	0
DisjointClasses	0
GCI count	0
Hidden GCI Count	0
Object property axioms	
SubObjectPropertyOf	166
EquivalentObjectProperties	0
InverseObjectProperties	0
DisjointObjectProperties	0
FunctionalObjectProperty	0
InverseFunctionalObjectProperty	0
TransitiveObjectProperty	0
SymmetricObjectProperty	0
AsymmetricObjectProperty	0
ReflexiveObjectProperty	0
IrrefexiveObjectProperty	0
ObjectPropertyDomain	152
ObjectPropertyRange	143
SubPropertyChainOf	0

Information Exchange Standard (IES)



Metrics	
Axiom	3990
Logical axiom count	1131
Declaration axioms count	714
Class count	513
Object property count	163
Data property count	43
Individual count	29
Annotation Property count	18
Class axioms	
SubClassOf	574
EquivalentClasses	0
DisjointClasses	0
GCI count	0
Hidden GCI Count	0
Object property axioms	
SubObjectPropertyOf	166
EquivalentObjectProperties	0
InverseObjectProperties	0
DisjointObjectProperties	0
FunctionalObjectProperty	0
InverseFunctionalObjectProperty	0
TransitiveObjectProperty	0
SymmetricObjectProperty	0
AsymmetricObjectProperty	0
ReflexiveObjectProperty	0
IrrefexiveObjectProperty	0
ObjectPropertyDomain	152
ObjectPropertyRange	143
SubPropertyChainOf	0

Information Exchange Standard (IES)





Metrics	
Axiom	3990
Logical axiom count	1131
Declaration axioms count	714
Class count	513
Object property count	163
Data property count	43
Individual count	29
Annotation Property count	18

Class axioms	
SubClassOf	574
EquivalentClasses	0
DisjointClasses	0
GCI count	0
Hidden GCI Count	0

Object property axioms	
SubObjectPropertyOf	166
EquivalentObjectProperties	0
InverseObjectProperties	0
DisjointObjectProperties	0
FunctionalObjectProperty	0
InverseFunctionalObjectProperty	0
TransitiveObjectProperty	0
SymmetricObjectProperty	0
AsymmetricObjectProperty	0
ReflexiveObjectProperty	0
IrrefexiveObjectProperty	0
ObjectPropertyDomain	152
ObjectPropertyRange	143
SubPropertyChainOf	0

Information Exchange Standard (IES)



Metrics

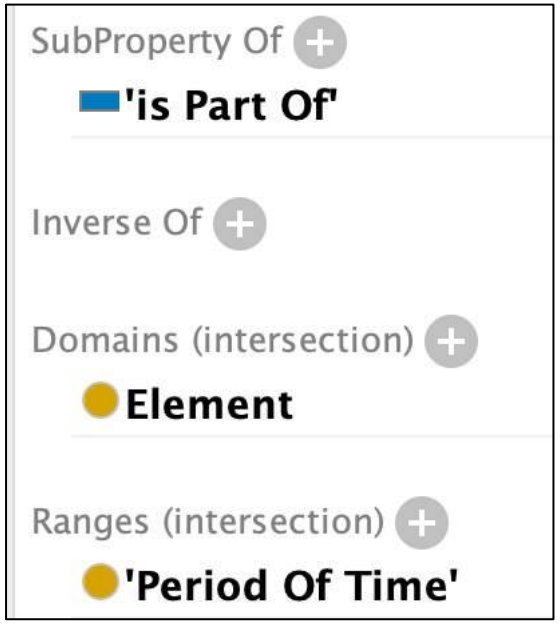
Axiom	3990
Logical axiom count	1131
Declaration axioms count	714
Class count	513
Object property count	163
Data property count	43
Individual count	29
Annotation Property count	18

Class axioms

SubClassOf	574
EquivalentClasses	0
DisjointClasses	0
GCI count	0
Hidden GCI Count	0

Object property axioms

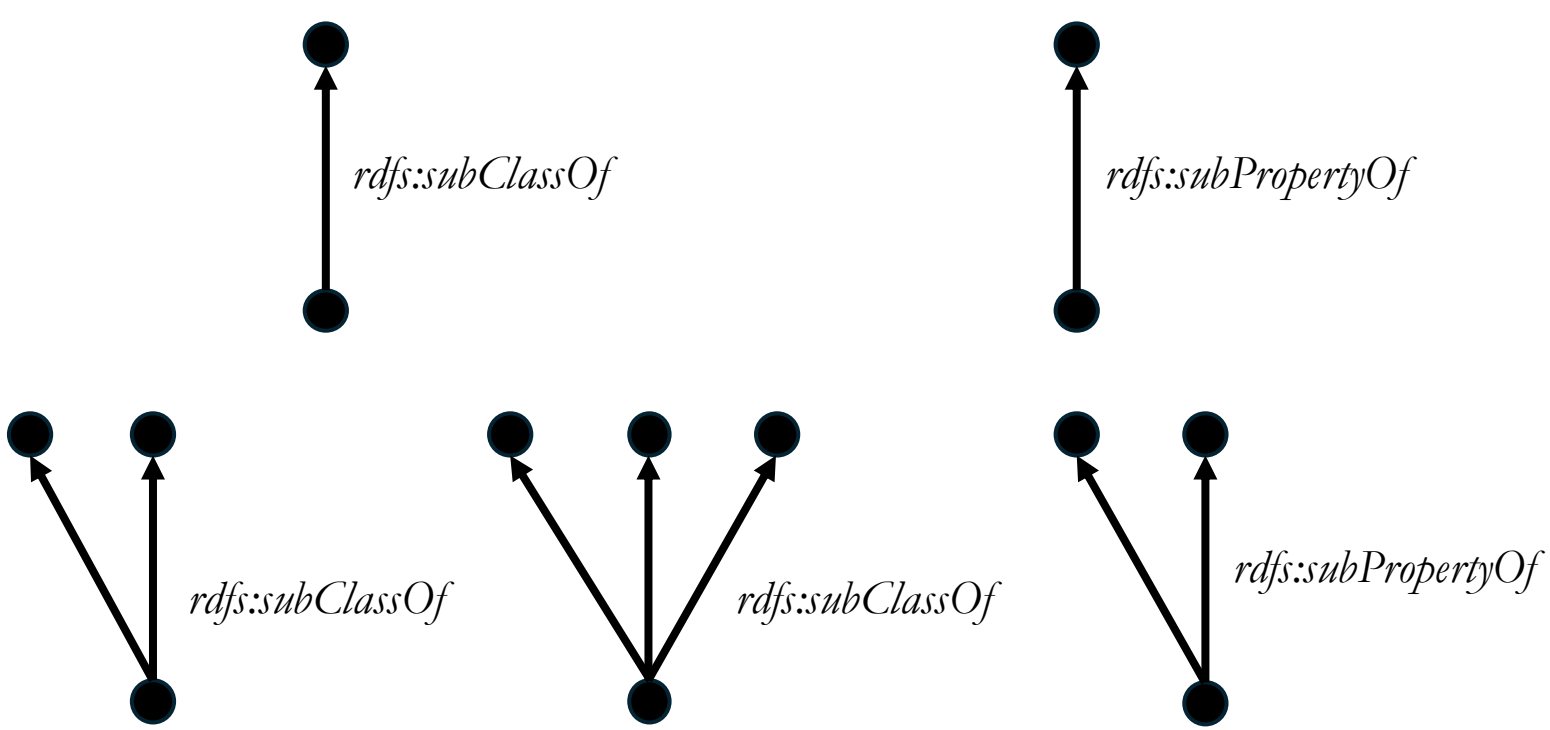
SubObjectPropertyOf	166
EquivalentObjectProperties	0
InverseObjectProperties	0
DisjointObjectProperties	0
FunctionalObjectProperty	0
InverseFunctionalObjectProperty	0
TransitiveObjectProperty	0
SymmetricObjectProperty	0
AsymmetricObjectProperty	0
ReflexiveObjectProperty	0
IrrefexiveObjectProperty	0
ObjectPropertyDomain	152
ObjectPropertyRange	143
SubPropertyChainOf	0



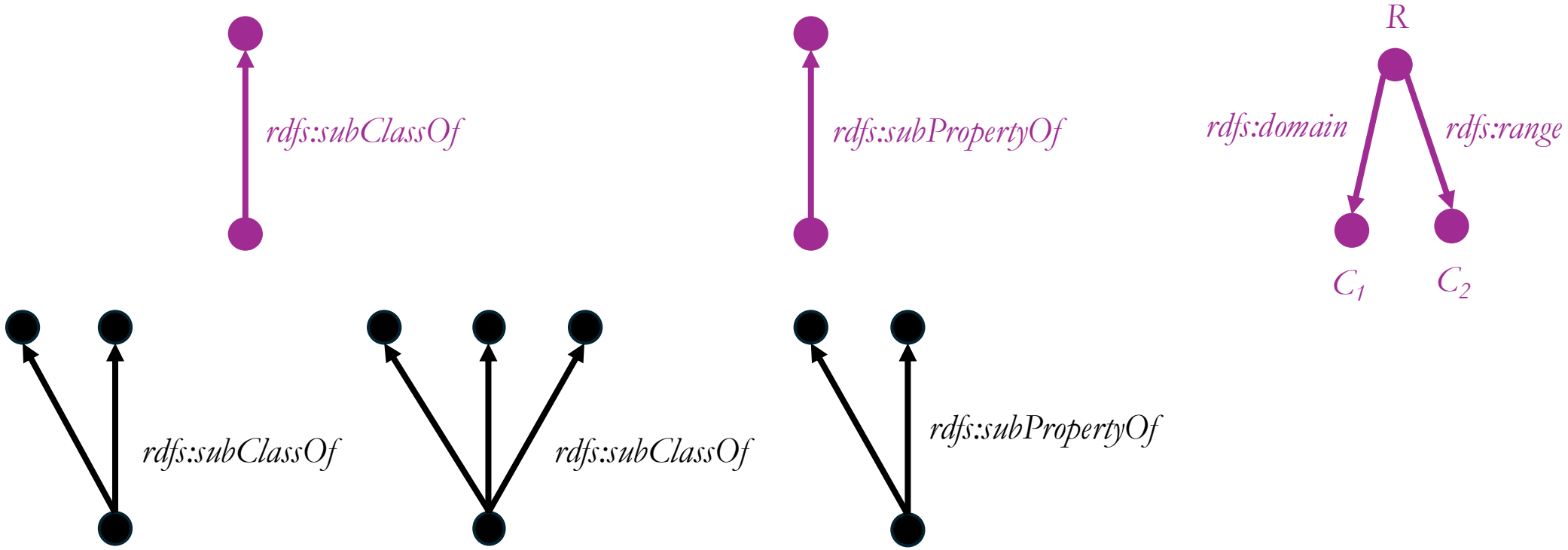
Axiom Patterns IES

- rdfs:subClassOf (allowing for polyhierarchy)
- rdfs:subPropertyOf (allowing for polyhierarchy)
- Domain/range restrictions

IES Axiom Patterns



Common Axiom Patterns



Challenge

- There must be more axiomatization added to IES before a mapping can be constructed

- On the one hand, any IES entity exibiting only subclass structure would be semantically equivalent to the subclass structure of any BFO entity

- On the other hand, because BFO exhibits richer axiomatization, very little in BFO will be semantically equivalent to IES

Challenge

- There must be more axiomatization added to IES before a mapping can be constructed

- On the one hand, any IES entity exibiting only subclass structure would be semantically equivalent to the subclass structure of any BFO entity

- On the other hand, because BFO exhibits richer axiomatization, very little in BFO will be semantically equivalent to IES

Challenge

- There must be more axiomatization added to IES before a mapping can be constructed

A GENERAL REMARK THOUGH ILLUSTRATED IN OWL

- On the one hand, any IES entity exibiting only subclass structure would be semantically equivalent to the subclass structure of any BFO entity

- On the other hand, because BFO exhibits richer axiomatization, very little in BFO will be semantically equivalent to IES

Example: Bank

An **Organisation** that **holds** a **banking license** and **can conduct financial transactions** on **behalf of customers**

Example: Bank

An **Organisation** that **holds** a **banking license** and **can conduct financial transactions** on **behalf of customers**

rdfs:subClassOf Organization

Example: Bank

An **Organisation** that **holds** a **banking license** and **can conduct financial transactions** on **behalf of customers**

???

Example: Bank

An **Organisation** that **holds** a **banking license** and **can conduct financial transactions** on **behalf of customers**

rdfs:subClassOf *holds license* some banking license

Example: Bank

An **Organisation** that **holds** a **banking license** and **can conduct financial transactions** on **behalf of customers**

rdfs:subClassOf ***holds license*** **some banking license and** ***authorized to conduct***

Example: Bank

An **Organisation** that holds a banking license and can conduct **financial transactions** on **behalf of customers**

rdfs:subClassOf *holds license* some banking license and *authorized to conduct* some (financial transaction

Example: Bank

An **Organisation** that holds a banking license and can conduct financial transactions on **behalf of customers**

rdfs:subClassOf *holds license* some banking license and *authorized to conduct* some (financial transaction **and** ***authorized by***

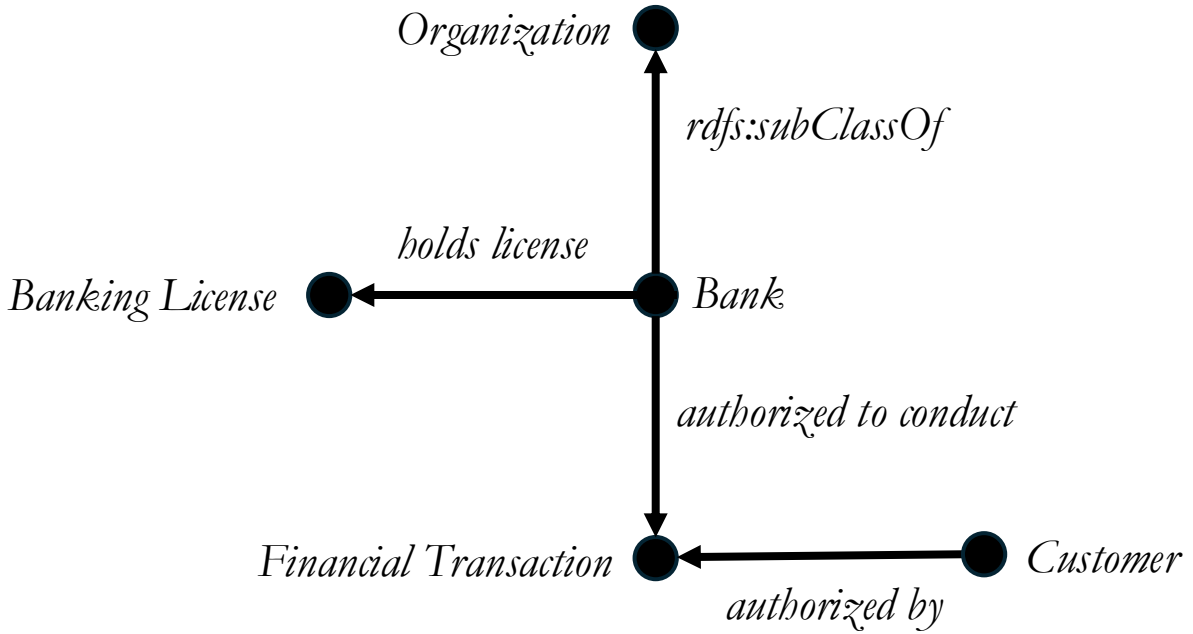
Example: Bank

An **Organisation** that holds a banking license and can conduct financial transactions on behalf of **customers**

rdfs:subClassOf *holds license* some banking license and *authorized to conduct* some (financial transaction and *authorized by* some **Customer**)

Example: Bank

rdfs:subClassOf *holds license* some banking license and *authorized to conduct* some (financial transaction and *authorized by* some Customer)

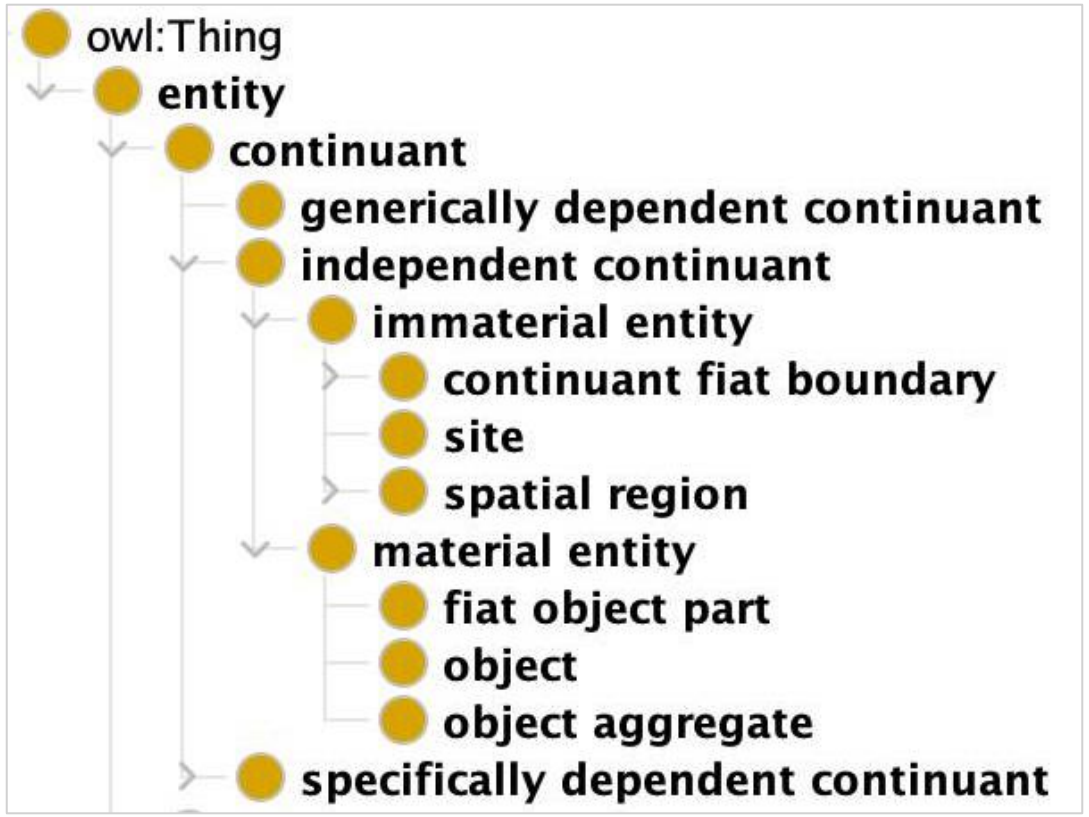


BUT JOHN - I HEAR YOU THINKING – WHAT IF THERE ARE FUNDAMENTAL DIFFERENCES BETWEEN ONTOLOGIES THAT MUST BE MAPPED?!?!?!?!

BUT JOHN - I HEAR YOU THINKING – WHAT IF THERE ARE FUNDAMENTAL DIFFERENCES BETWEEN ONTOLOGIES THAT MUST BE MAPPED?!?!?!?!

REST EASY, LOGIC PROVIDES A PATH FORWARD

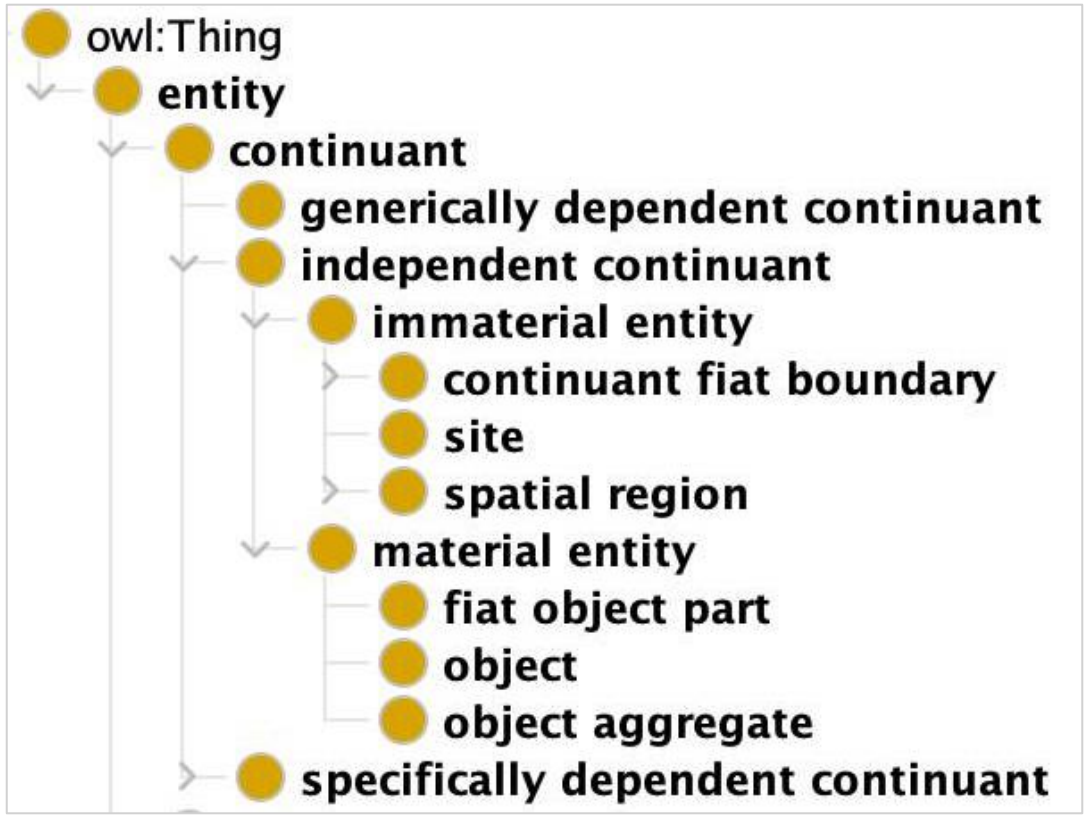
Generating Implicit Hierarchy





Suppose there is a need to maintain an application ontology that is not aligned to BFO

Generating Implicit Hierarchy

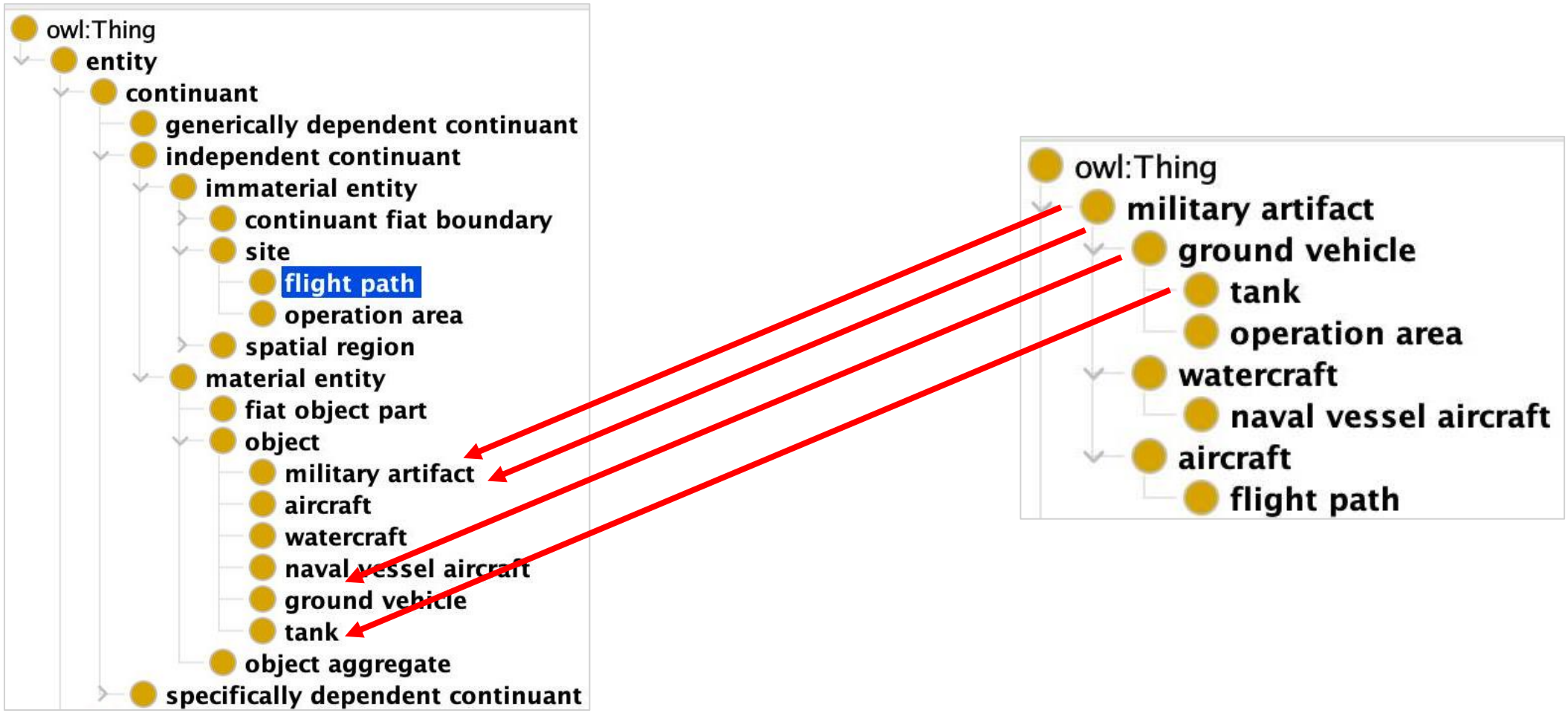


≠

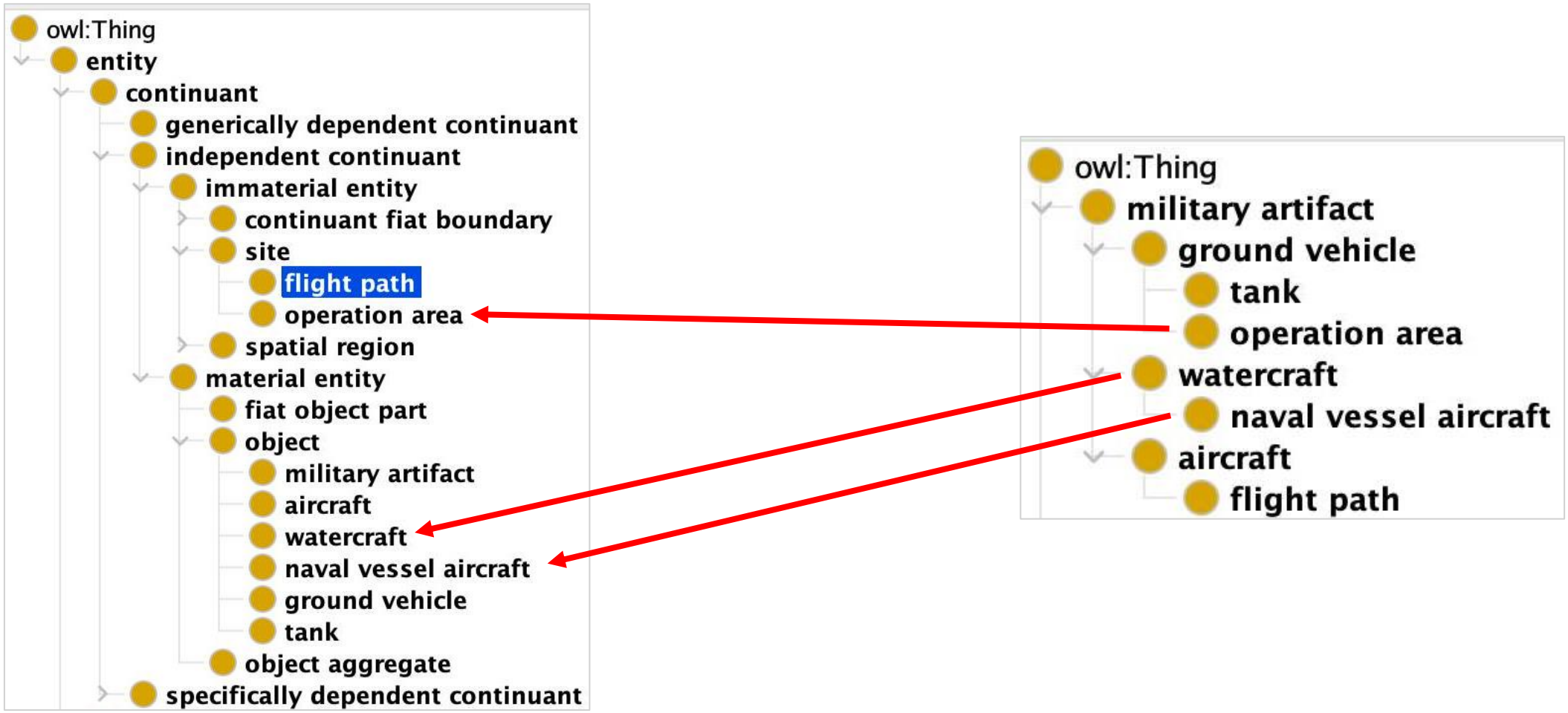


Suppose there is a need to maintain an application ontology that is not aligned to BFO

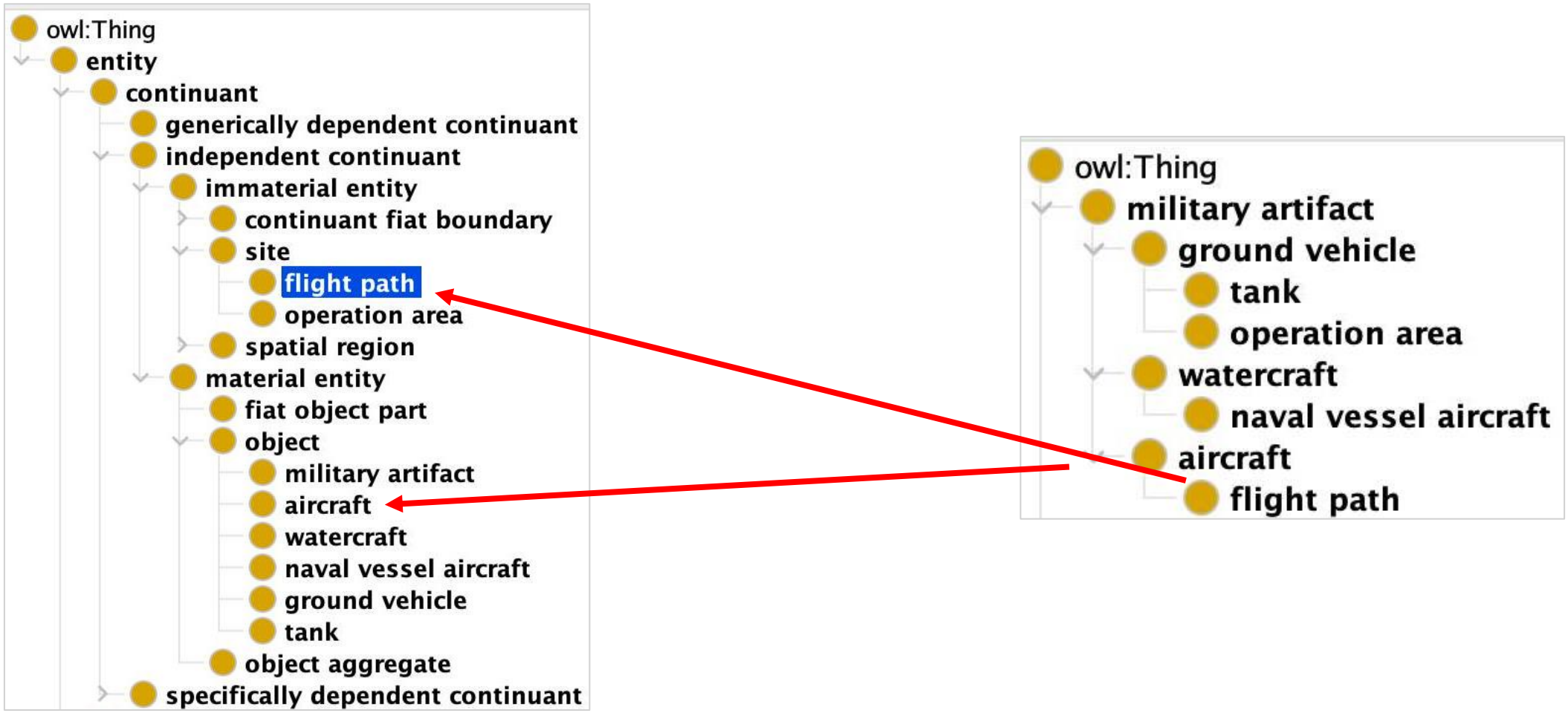
Generating Implicit Hierarchy



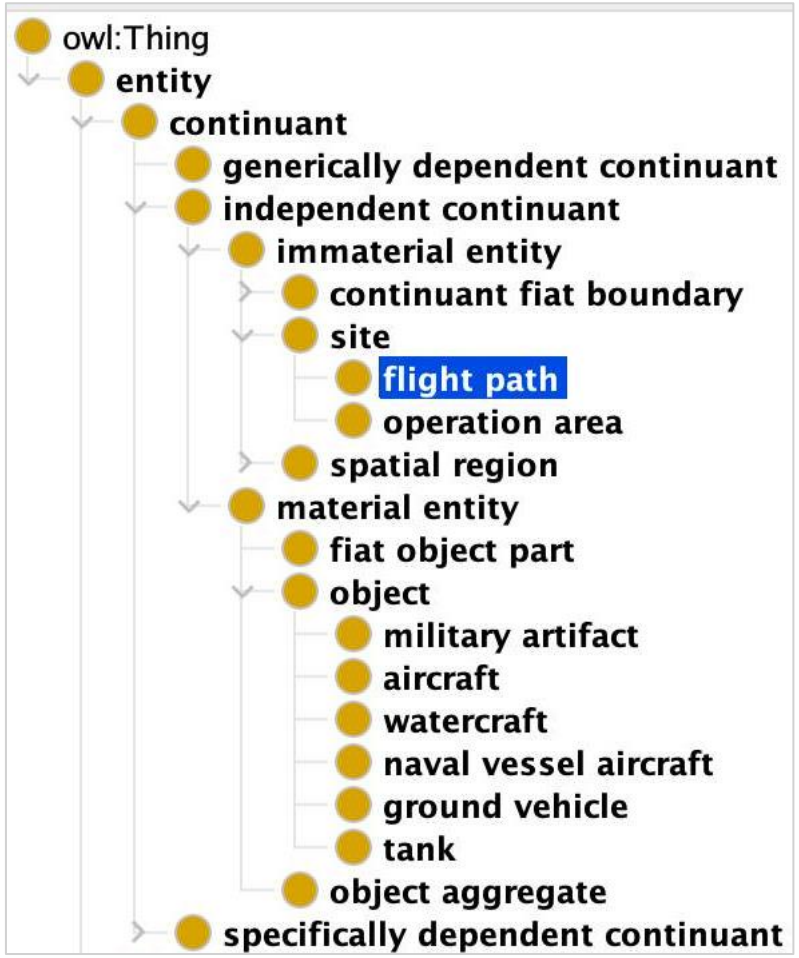
Generating Implicit Hierarchy



Generating Implicit Hierarchy



Generating Implicit Hierarchy



≠



military taxonomy relation

- One way forward is to introduce an object property – call it **military taxonomy** – that connects instances of the reference ontology to those of the military artifact class

- To simulate the **subclass of** relation, we assert that **military taxonomy** is **reflexive** and **transitive**

- And is such that any entity in the domain can be related **only** to instances under military artifact

owl:allValuesFrom

- All instances that are related to **only** some C

THING

{<x,y> | <x,y> ∈ R implies y ∈ C}

owl:ReflexivityProperty

- For all x, x is related to x

THING

$$\{\langle x,x\rangle \mid x \in \text{Domain}\} \subseteq R^I$$

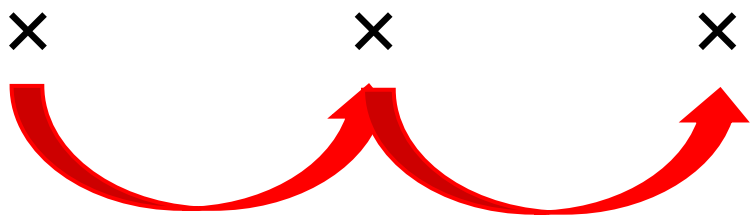


owl:TransitiveObjectProperty

- If x related to y and y related to z, then x related to z

THING

$$\mathbf{Trans(R) = R^I \circ R^I \subseteq R^I}$$

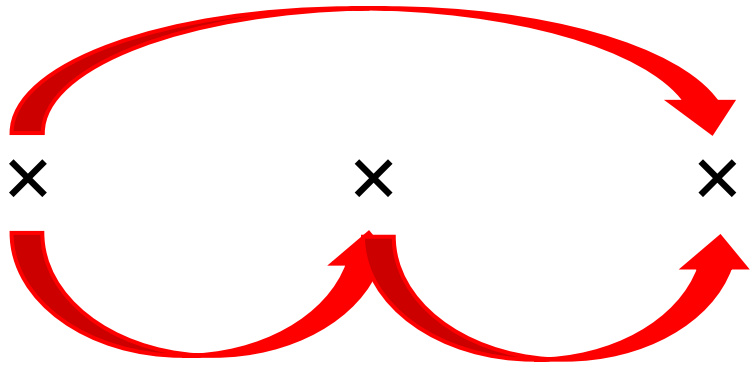


owl:TransitiveObjectProperty

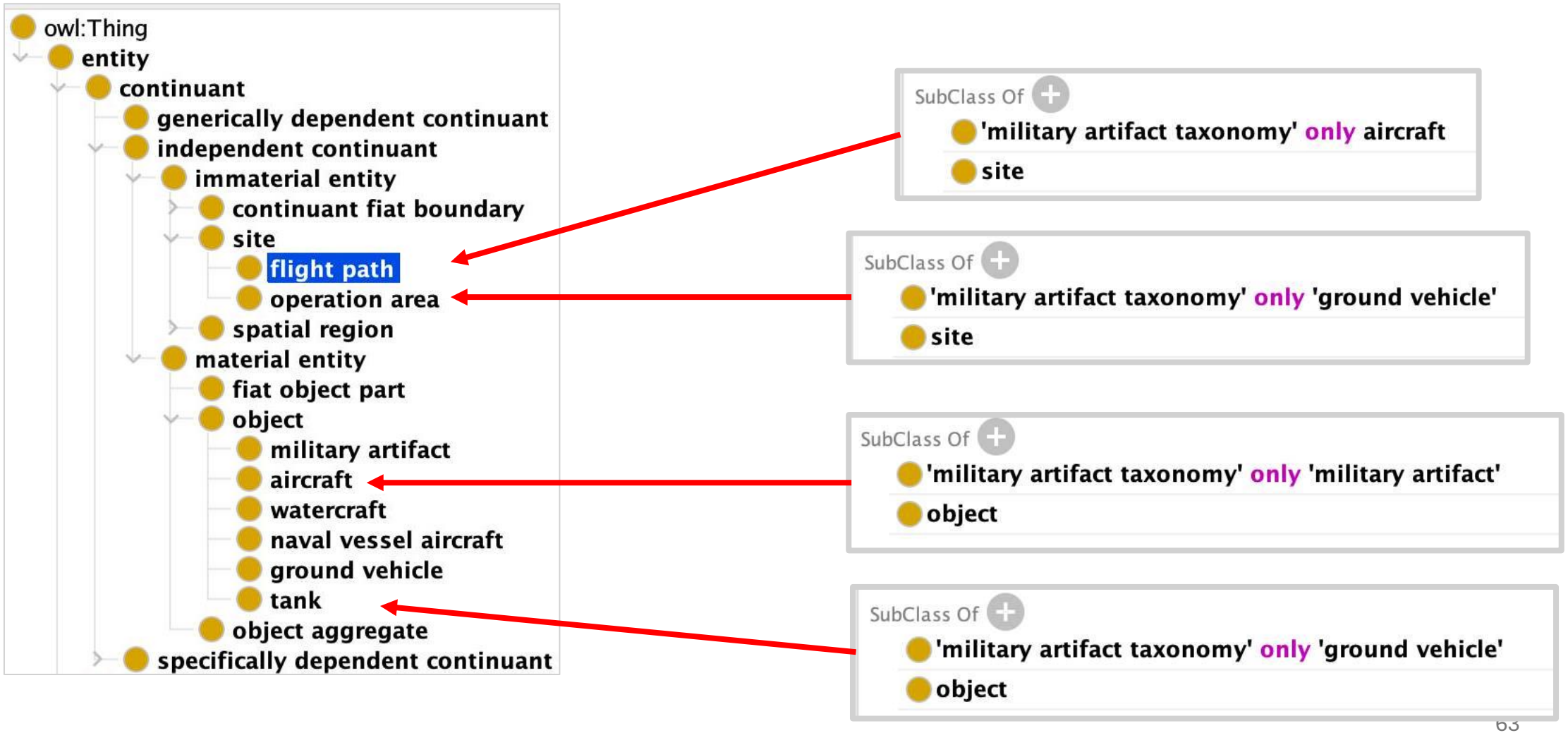
- If x related to y and y related to z, then x related to z

THING

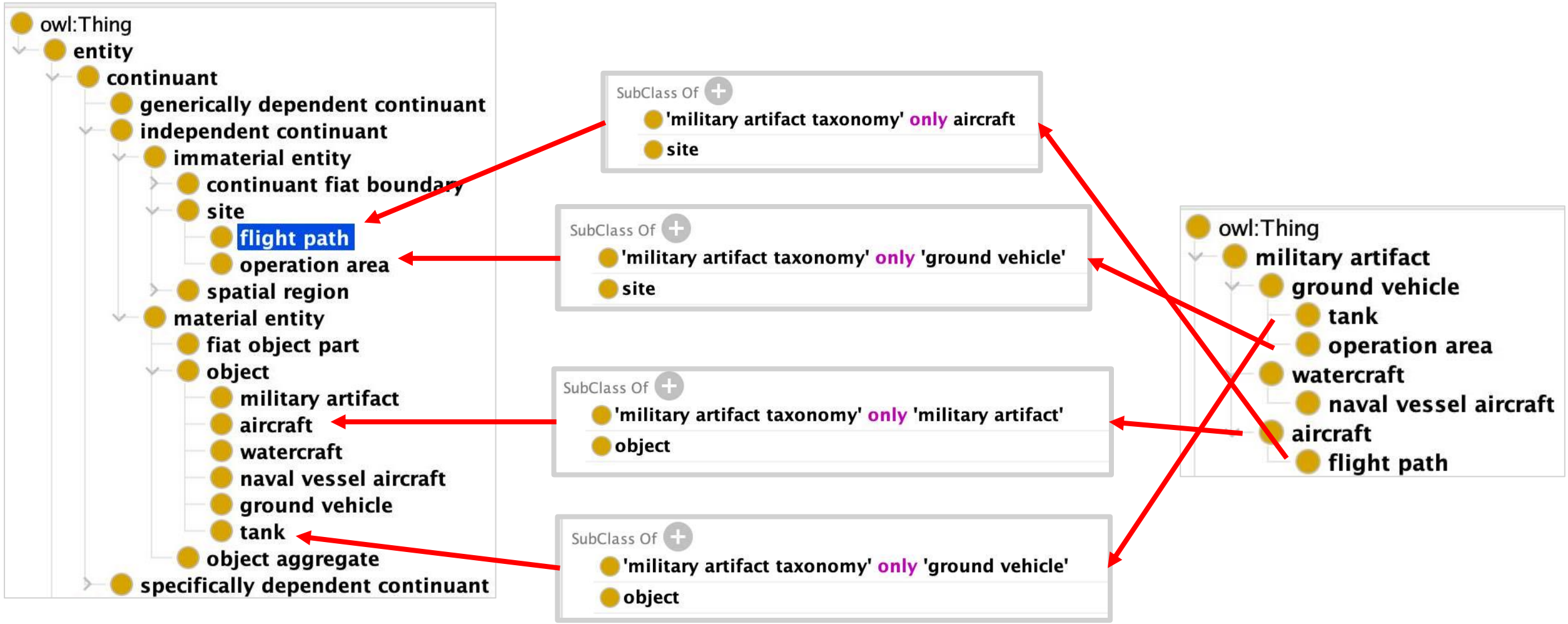
$$\mathbf{Trans(R) = R^I \circ R^I \subseteq R^I}$$



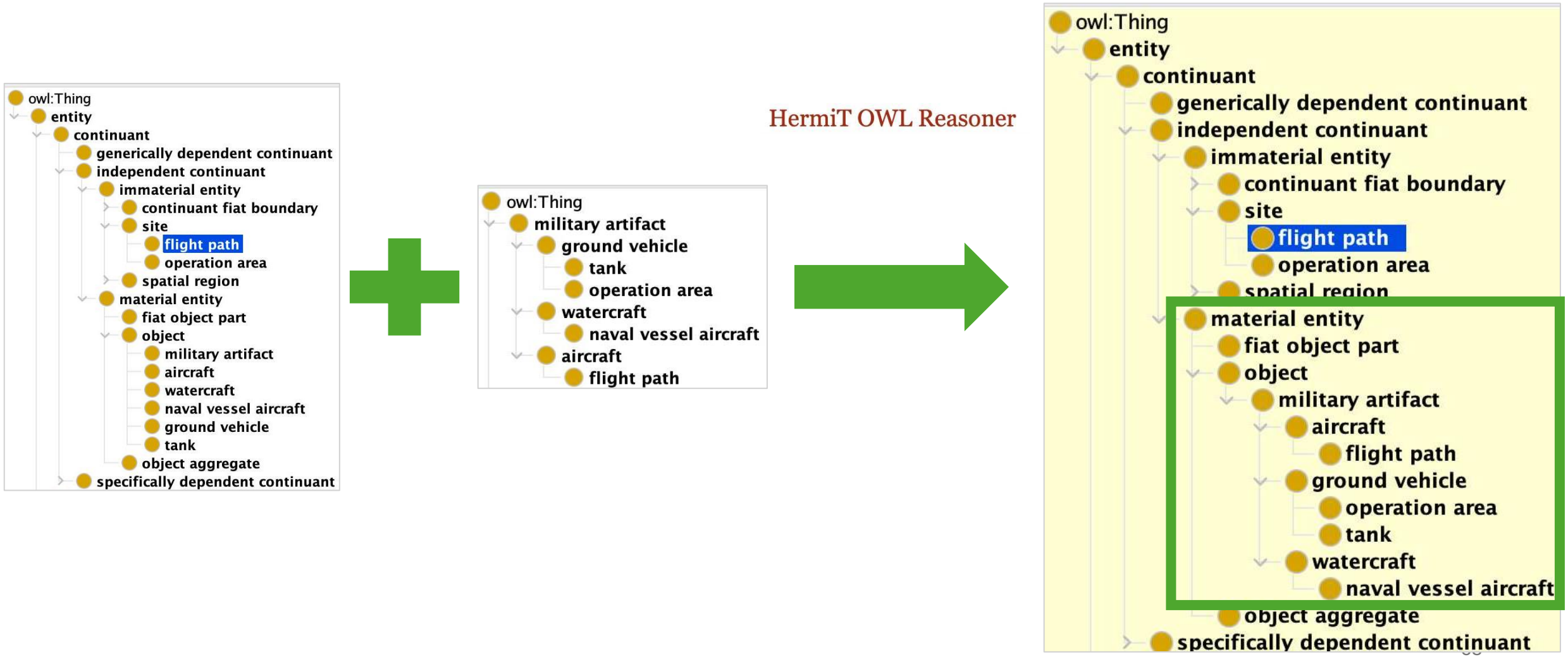
Generating Implicit Hierarchy



Generating Implicit Hierarchy



Generating Implicit Hierarchy

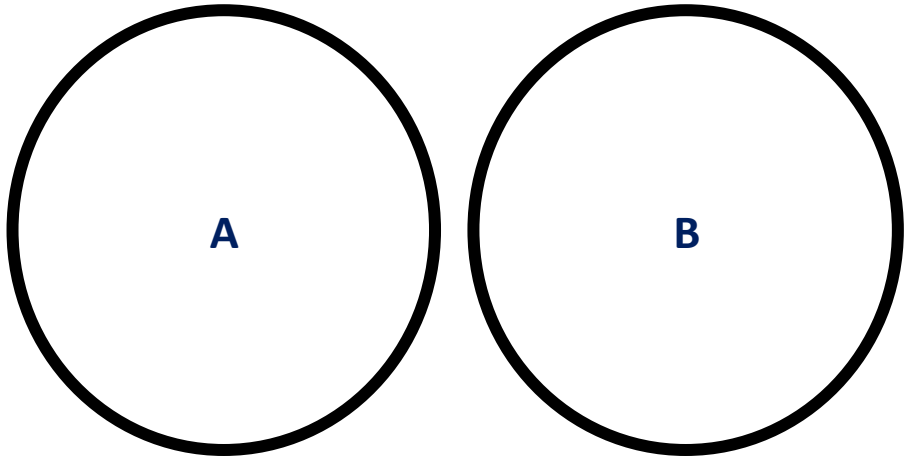


Caveat: Disjointness

- Disjointness – A and B are disjoint just in case they share no individuals

THING

$$\mathbf{DisjointWith(A, B) = A^I \cap B^I = \emptyset}$$



Caveat: Disjointness Must be Dropped

- In BFO, the class **site** and the class **object** are **disjoint**, which means they may share no instances in common

- Consequently, **operation area** cannot – strictly speaking – be an asserted subclass of **object** and inferred subclass of **site**

- Importantly, such constraints should be understood as applying at the level of reference ontologies, **not necessarily** application ontologies

Mapping Files

- Mapping files need not abide by the principles governing ontology design, insofar as those principles undermine mission goals

- When principles are not adhered to owing to application needs, keep the new application ontology distinct from relevant reference ontologies and **annotate deviations from reference ontology in detail**

ANNOTATIONS

Guiding Questions

How much alignment can you enforce across communities without undermining user goals?

What can be done to minimize downstream alignment efforts across diverging communities?

Guiding Questions

How much alignment can you enforce across communities without undermining user goals?

What can be done to minimize downstream alignment efforts across diverging communities?

What can be done with ontology engineering in application?

Learning Goals

By the end of the exercise, you should be able to:

1. Explain the difference between deductive and abductive reasoning.
2. Encode puzzle constraints as OWL axioms.
3. Use a generative AI model to make abductive guesses from incomplete information.
4. Use HermiT to determine what follows deductively from an OWL ontology.
5. Evaluate how the outputs of a generative AI model change as additional axioms are supplied.

What You Will Need

- Protégé and the HermiT reasoner
- Access to a generative AI model of your choice
- A copy of this repository

Important Rule for the AI Model

The Zebra Puzzle is widely known, and its solution is available online. Tell the model:

> Do not search the web, retrieve a memorized solution, or rely on prior knowledge of the Zebra Puzzle. Reason only from the clues supplied in this prompt.

The purpose of the exercise is to observe abductive reasoning from incomplete information. Looking up the known solution undermines the experiment.

Uses of Formal Methods

- For some interpretations T, formulas p, and models M:

- Is a fixed p true **for all T**?
- Is a fixed p true **for some T**?
- For fixed M **what p makes M true**?
- For fixed M and p **does M model p**?

Uses of Formal Methods

- For some interpretations T, formulas p, and models M:

- Is a fixed p true **for all T**?
- Is a fixed p true **for some T**?
- For fixed M **what p makes M true**?
- For fixed M and p **does M model p**?

Is "2+2=4" always true?

Uses of Formal Methods

- For some interpretations T, formulas p, and models M:

- Is a fixed p true **for all T**?
- Is a fixed p true **for some T**?
- For fixed M **what p makes M true**?
- For fixed M and p **does M model p**?

Is "2+2=4" always true?

Can "John is bald" be true?

Uses of Formal Methods

- For some interpretations T, formulas p, and models M:

- Is a fixed p true **for all T**?
- Is a fixed p true **for some T**?
- For fixed M **what p makes M true**?
- For fixed M and p **does M model p**?

Is "2+2=4" always true?

Can "John is bald" be true?

Build traffic light with alternating light color.

Uses of Formal Methods

- For some interpretations T, formulas p, and models M:

- Is a fixed p true **for all T**? **Is "2+2=4" always true?**
- Is a fixed p true **for some T**? **Can "John is bald" be true?**
- For fixed M **what p makes M true**? **Build traffic light with alternating light color.**
- For fixed M and p **does M model p**? **Does this traffic light alternate colors?**

Uses of Formal Methods

- For some interpretations T, formulas p, and models M:

- Is a fixed p true **for all T**? **Proof Checking/Assistants**
- Is a fixed p true **for some T**? **SAT Solving**
- For fixed M **what p makes M true**? **Model Synthesis**
- For fixed M and p **does M model p**? **Model Checking**

Question	Example	Strategy
Is a fixed proposition *p* true for all interpretations *T*?	Given the rules **"All humans are mortal"** and **"Socrates is human,"** is **"Socrates is mortal"** necessarily true?	**Automated Theorem Proving** Establish that the conclusion follows in every interpretation.
Is a fixed proposition *p* true for some interpretation *T*?	Can (p = A & ~B) be true for **some** assignment of truth values? Yes: (A=true, B=false).	**SAT Solving** Find at least one assignment/model satisfying the formula.
For fixed model *M*, what proposition *p* makes *M* true?	Suppose the requirement is **"Exactly one of Alice or Bob owns the zebra."** Find an assignment such as **Alice owns it, Bob does not** that satisfies the requirement.	**Model Synthesis** Construct a model or assignment of T and F satisfying the given constraints.
For fixed model *M* and proposition *p*, does *M* model *p*?	In a particular model, **Alice owns the zebra**. Does that model satisfy (p=) **"Someone owns the zebra"**? Yes.	**Model checking:** take a particular model and test whether a particular statement holds in it.

Uses of Formal Methods

- For some interpretations T, formulas p, and models M:

- Is a fixed p true **for all T**? **Proof Checking/Assistants**
- Is a fixed p true **for some T**? **SAT Solving**
- For fixed M **what p makes M true**? **Model Synthesis**
- For fixed M and p **does M model p**? **Model Checking**

Formal Verification

- *Formal verification* involves constructing an **intended model** of some system architecture, such as hardware, software, communication protocols, business logic, etc...

- ...described by a **specification** in a formal language...

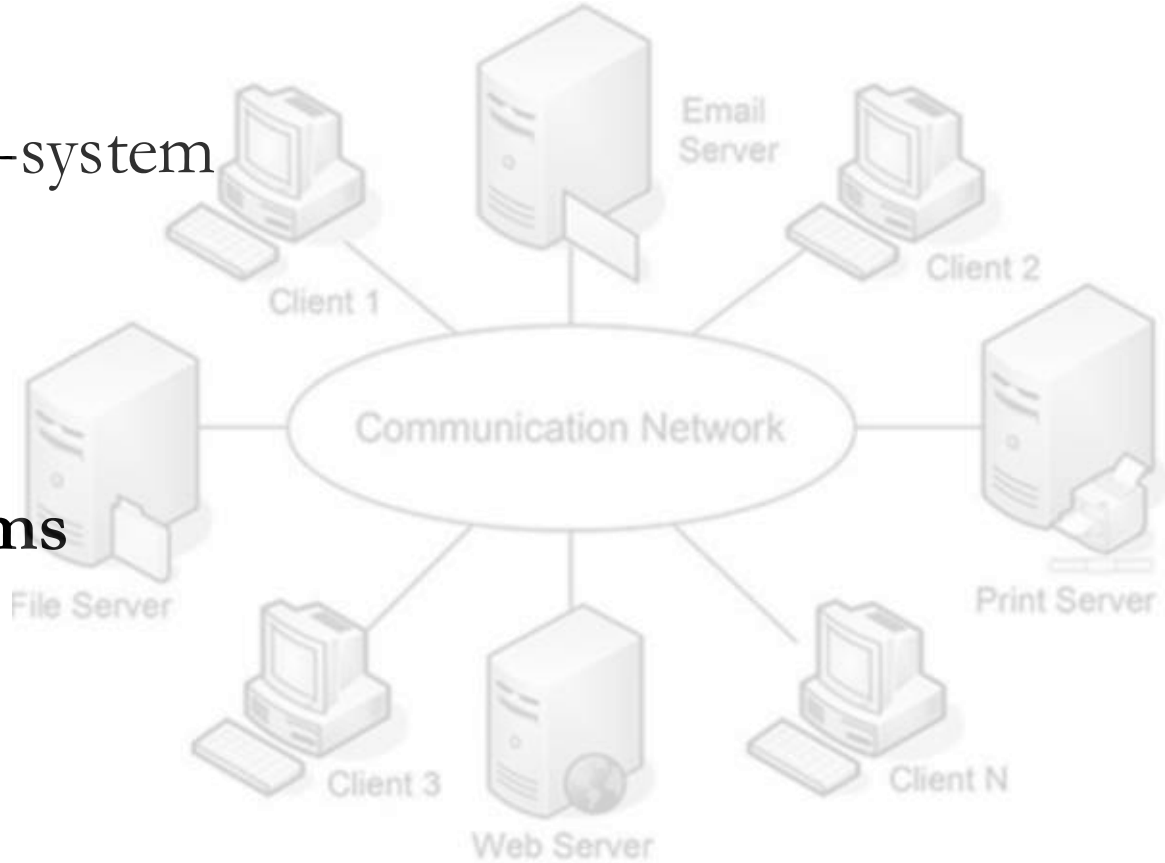
- ...so that automated model checking tools can be used to determine whether a **formal model** of the **specification** satisfies the **intended model** of the architecture

Distributed System

- *Distributed System* – Connected sub-systems lacking access to a global memory

- For distributed systems, failure of a sub-system you don't even know exists, can render your local sub-system unusuable.

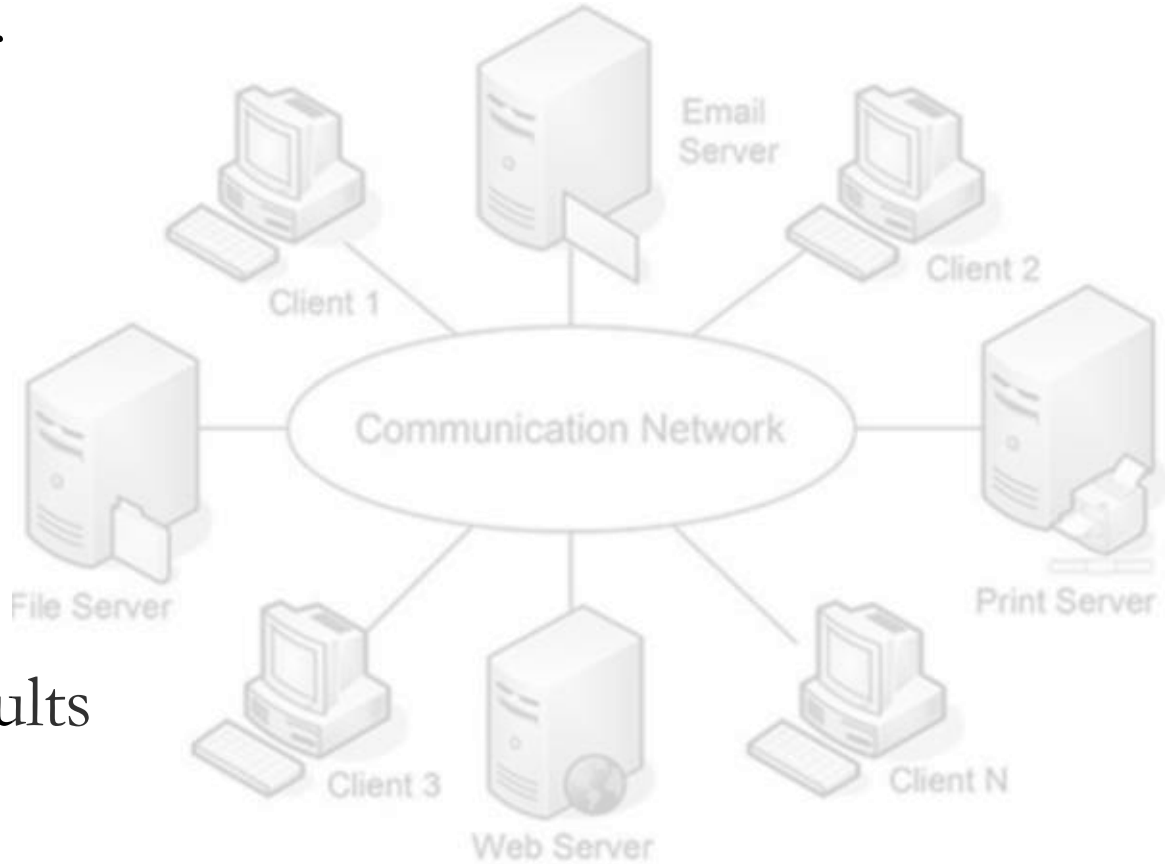
- **"Everything about distributed systems is terrible" – Hillel Wayne**



Distributed System Faults

- *Faults* – Disruptions in expected behavior classified as:
  - **Permanent** – Persists until corrected
  - **Intermittent** – Occur, vanish, occur
  - **Transient** – Occur, self-correct

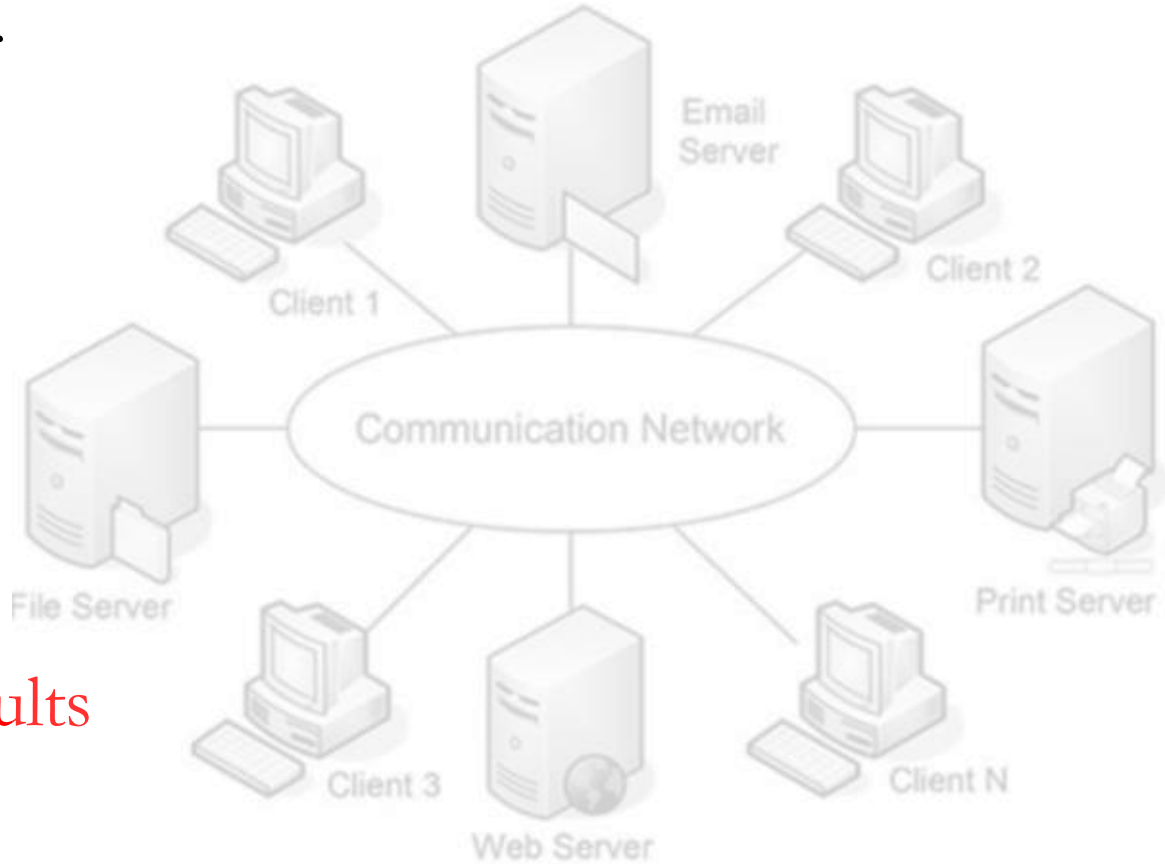
- Any of these can be:
  - **Fail-Silent** – Failure produces no bad output
  - **Byzantine** – Continues to run but produces incorrect results



Distributed System Faults

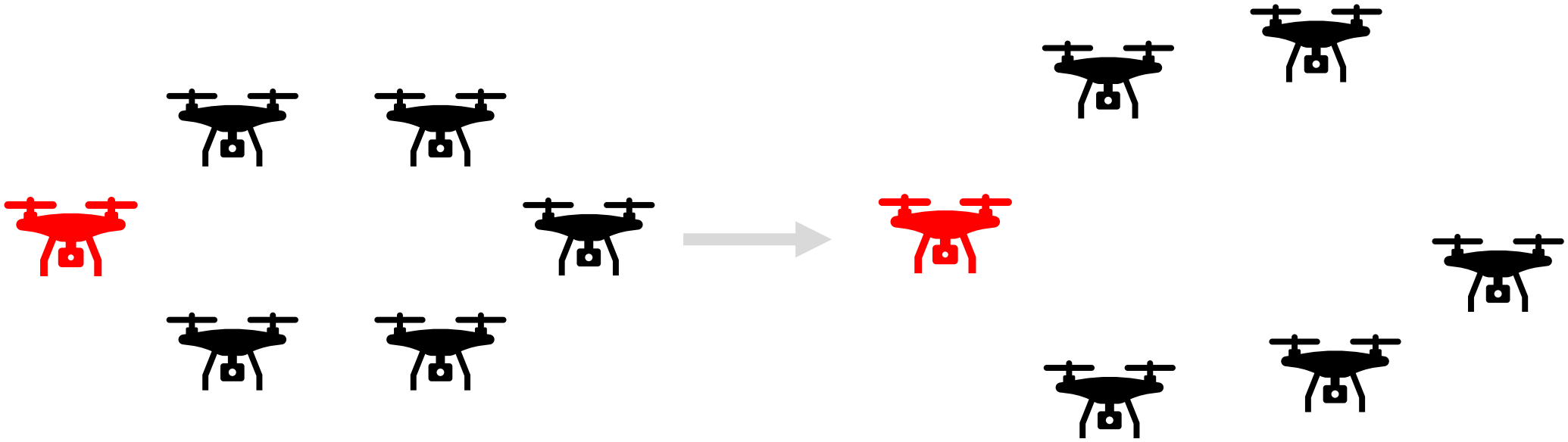
- *Faults* – Disruptions in expected behavior classified as:
  - **Permanent** – Persists until corrected
  - **Intermittent** – Occur, vanish, occur
  - **Transient** – Occur, self-correct

- Any of these can be:
  - **Fail-Silent** – Failure produces no bad output
  - **Byzantine** – Continues to run but produces incorrect results



Byzantine Failure

Drone swarm in coordinated flight. Each drone sends relative position data to others. Suppose Drone A experiences a Byzantine failure such that it sends inaccurate, inconsistent, data to other drones. For example, the Drone A sends "I am ascending"to drone B and "I am descending" to Drone C.



Model Checking Rescue

- Can be used to represent **all possible states of the system**; byzantine faults can be identified in the design stage.

- A **specification** of the system will consist of:
  - A finite set of states.
  - A subset that includes an initial state.
  - A subset that includes an ending state.
  - A transition relation that describes all states that can be reached incrementally.

Specification Properties to Check

- **Safety** – Propositions ensuring invalid states are not reachable.
  - **Deadlock** – Proposition ensuring no dead end state is reached.

Violation of Deadlock: Process A requires resource from Process B before acting and Process B requires resource from Process A

Specification Properties to Check

- **Liveness** – Propositions reflecting states that should occur.
  - **Termination** – Proposition ensuring end state is eventually reached.
  - **Weak Fairness** – If action remains enabled it will eventually occur.
  - **Strong Fairness** – If action is repeatedly enabled it will eventually occur.

Violations of Termination: System crash; infinite loop

Violation of Weak Fairness: Whenever process A is enabled, Process B blocks transition.

Violation of Strong Fairness: Whenever process A is enabled, Process B disables it.

Future of Research and Development

- Traditional ontology engineering now lives at the intersection of **policy**, **governance**, and **integration** in the wider software stack

- The future will see emphasis on **robust semantic mappings** within and across ontology ecosystems, via executable transforms which themselves require great rigor

- Emphasis will be placed on **identifying the best propositions or questions** to pose to a well-developed ontology ecosystem

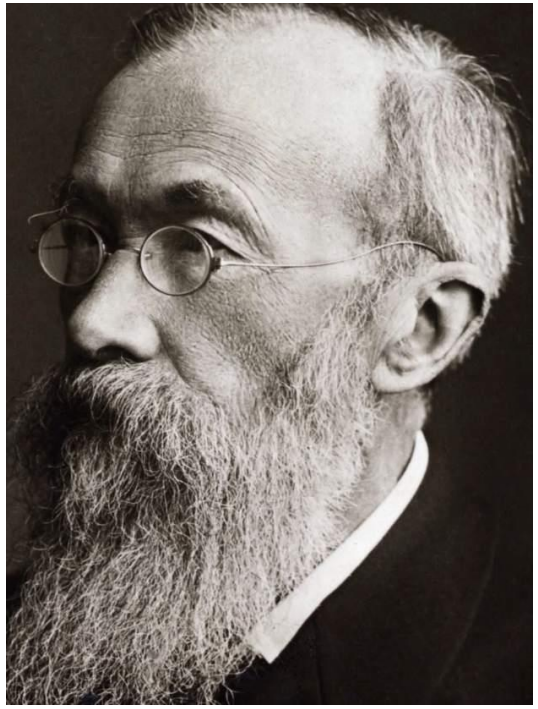
Outline

- Research and Development
- The Structure of Scientific Revolutions
- A New Era

Garden of Forking Paths

- Philosophy is a garden in which scientific disciplines are grown







Garden of Forking Paths

- Philosophy is a garden in which scientific disciplines are grown
- Ontology engineering – despite existing as a field for over 20 years – has yet to fully emerge as a scientific discipline
- Philosophers turned engineers and engineers turned philosophers **do not** a discipline make
- We need **consensus on methods**

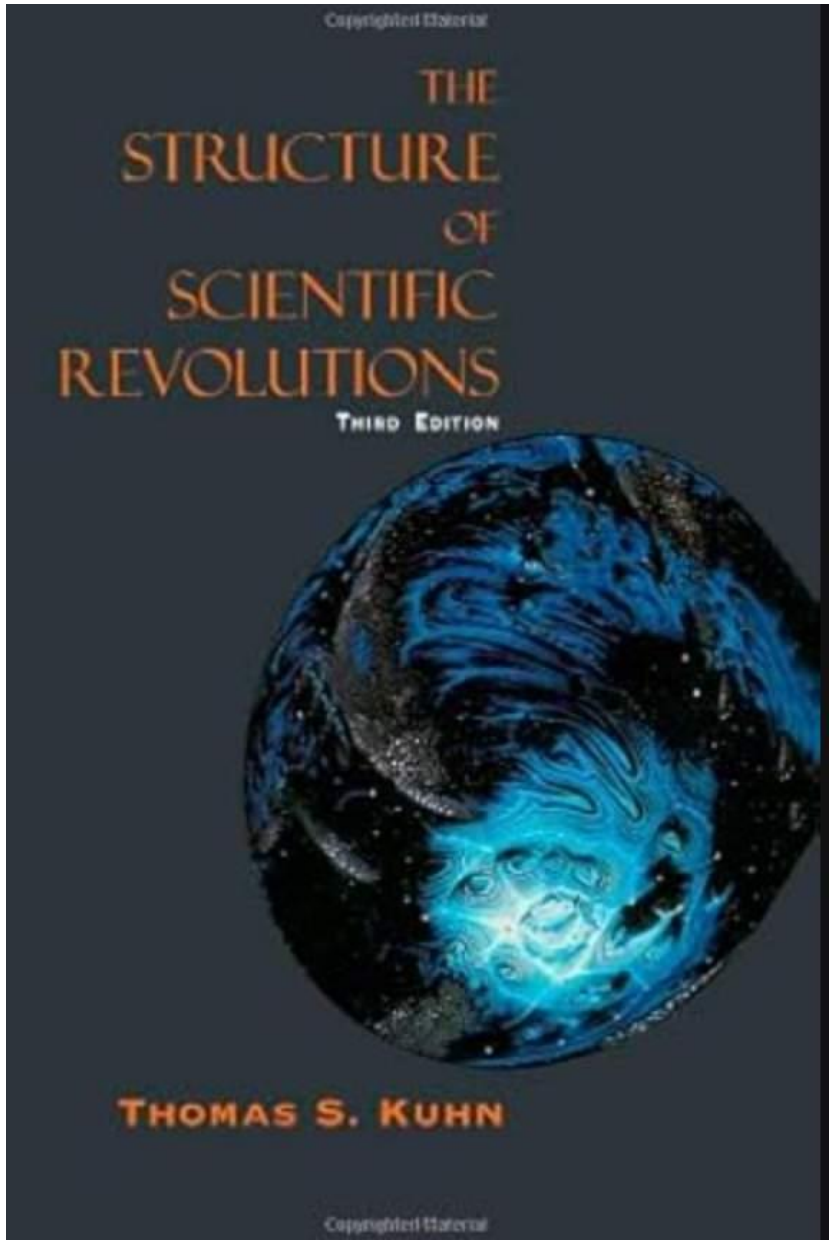
Ontology engineering is focused on constructing machine-interpretable artifacts designed to systematically disambiguate information, improve information quality, and facilitate information interoperability

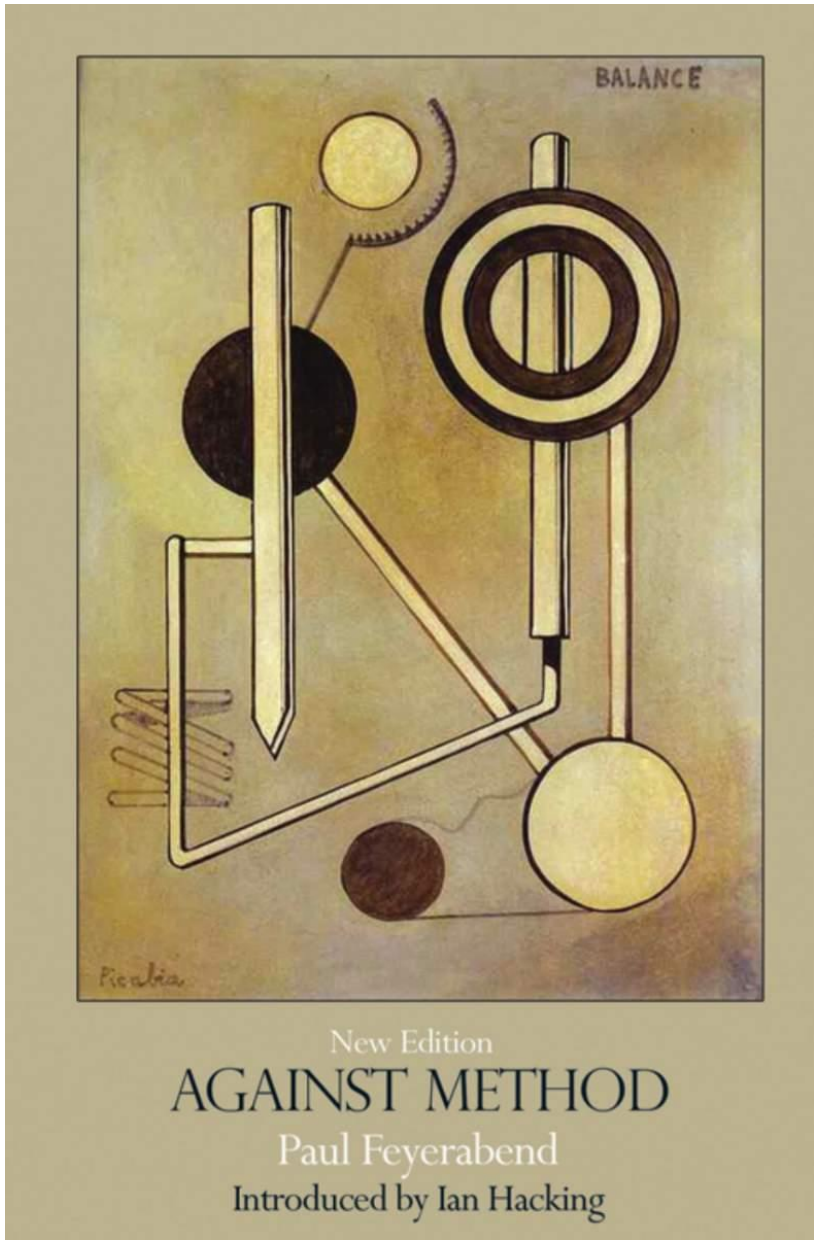
WE, OF COURSE, NEED MORE THAN AGREEMENT

WE, OF COURSE, NEED MORE THAN AGREEMENT



national
center for
ontological
research
ncor
academy





Dissemination

national center for ontological research
ncor
academy

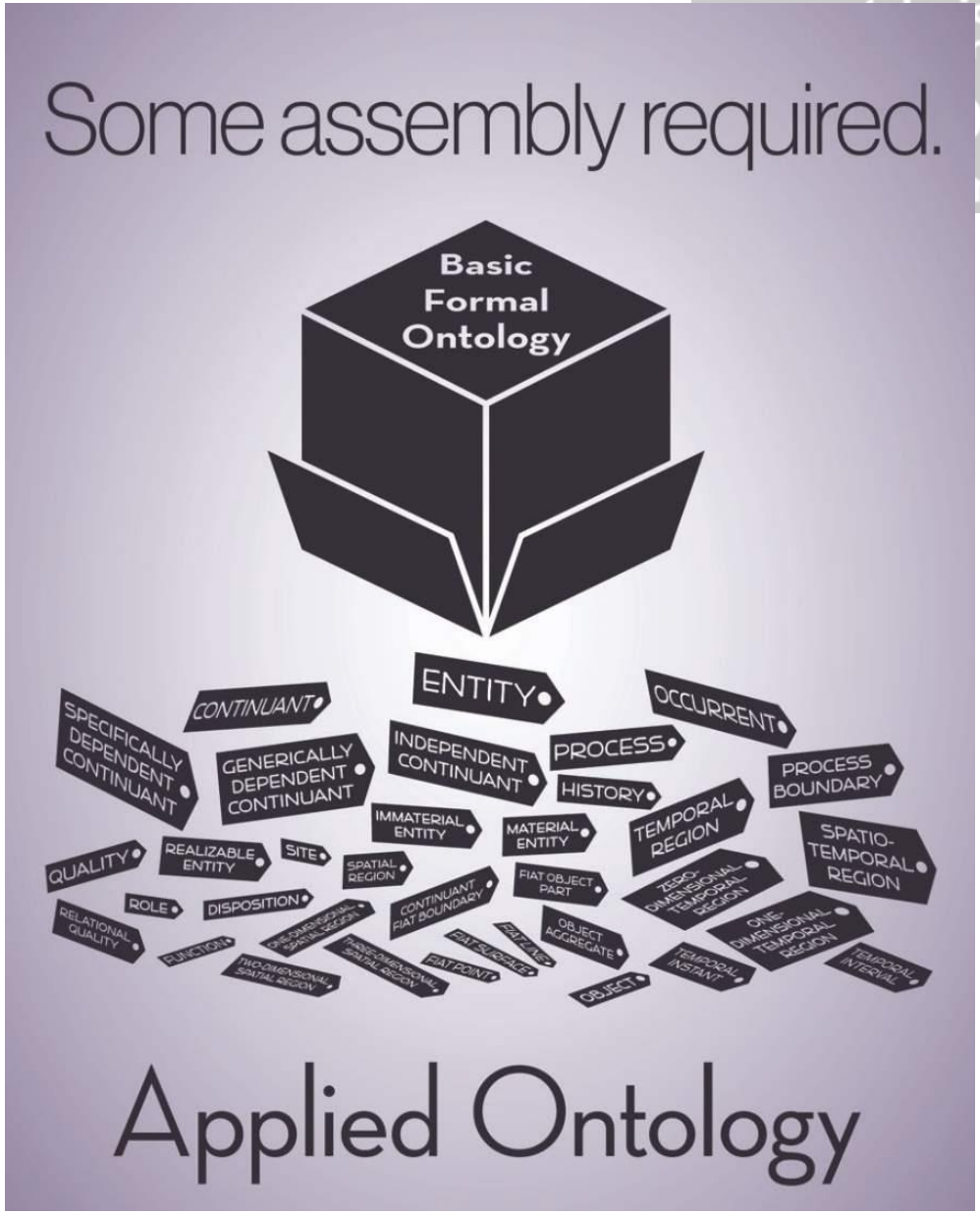
- This is not a **mere intellectual or academic exercise**; the position has and will continue to impact **training**, **research**, and **application**

- E.g. Barry and I have (re)trained **numerous ontology engineers** across government, academic, and commercial spaces in **the last three years**

- If you disagree, or think the proposal should be amended, speak up sooner rather than later **because**...

UB Applied Ontology

- In Fall 2026, UB will enroll the first class of applied ontology PhD students in the US, and perhaps the world
- At present count, we have a class of ~50 students across MS/PhD
- The program is **entirely remote** and **asynchronous**



Overview | Repositories 9 | Projects | Packages | Teams | People 1 | Insights | Settings

national center for ontological research
ncor
academy

Applied Ontology Education

Organization for applied ontology education

Unfollow

32 followers | United States of America | https://ncor-organization.github.io/... | https://www.youtube.com/c/BarryS... | @johnbeve

https://ontology.buffalo.edu/smith/ | https://johnbeverley.com/ | johnbeve@buffalo.edu

README.md

Applied Ontology Education GitHub Organization

Welcome to the official NCOR GitHub organization for ontology engineering educational materials. The resources contained here are designed to guide you through the evolving fields of semantic web technologies, AI-powered systems, and advanced data modeling.

Our Mission Our goal is to empower learners, developers, and researchers in their journey towards mastering ontology engineering, knowledge graph development, and artificial intelligence. By providing curated resources and hands-on repositories, we hope to facilitate knowledge sharing and skill-building within these important domains.

View as: Public

You are viewing the README and pinned repositories as a public user.

You can pin repositories visible to anyone.

Get started with tasks that most successful organizations complete.

Discussions

Set up discussions to engage with your community!

Turn on discussions

101 Working Group





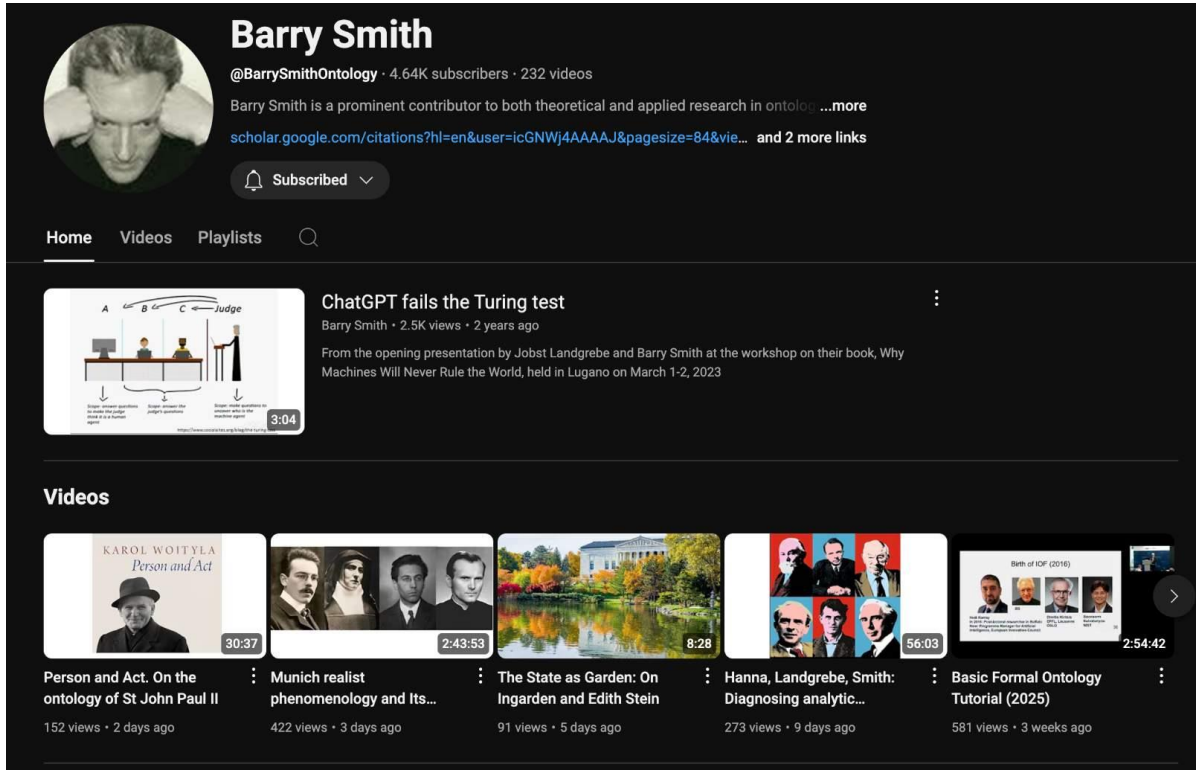
- Chairs: Bill Mandrick & John Beverley

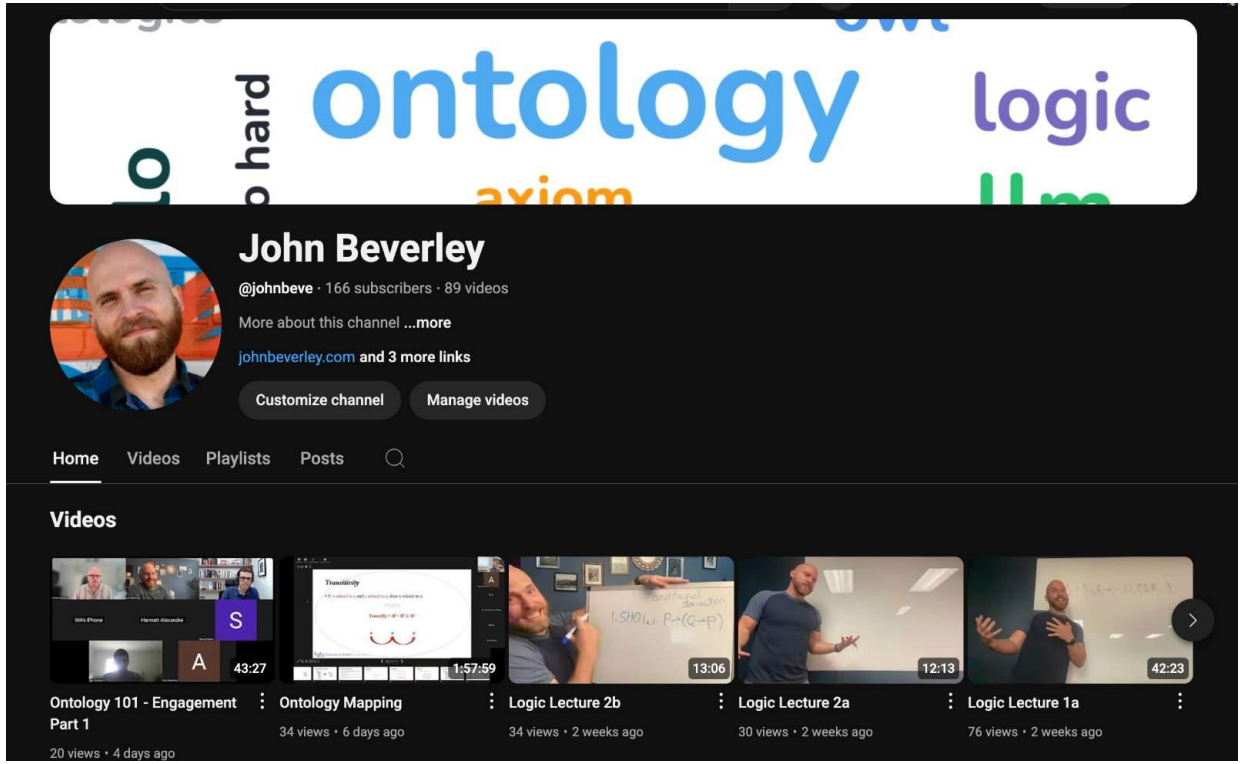
- 6-Week Cycle Format:
  - **Subject Matter Expert (SME) Engagement** – Practice eliciting competency questions, handling objections, and navigating challenging SME dynamics.
  - **Design Pattern Construction** – Model those questions into reusable visual patterns with critique and iteration.
  - **OWL Representation** – Translate your patterns into formal OWL in Protégé and test your models with reasoners.

- Weekly meetings at **11am on W**; next cycle starts **Sept 3**

MIT Model

- All of our education material will remain accessible online: https://github.com/Applied-Ontology-Education





NCOR 501(c)3



- Established as a non-profit at the end of 2024

- Board of Directors: Barry Smith & John Beverley

- Funding:
  - Donations
  - Overflow from co-Hosting STIDS 2024 with GMU & KadSci
  - NIST Sustainability Grant with Crownpoint Technologies
  - Air Combat Command US Air Force
  - Acacia KS submitted AF grants with WTI; establishing JV for future collaborations

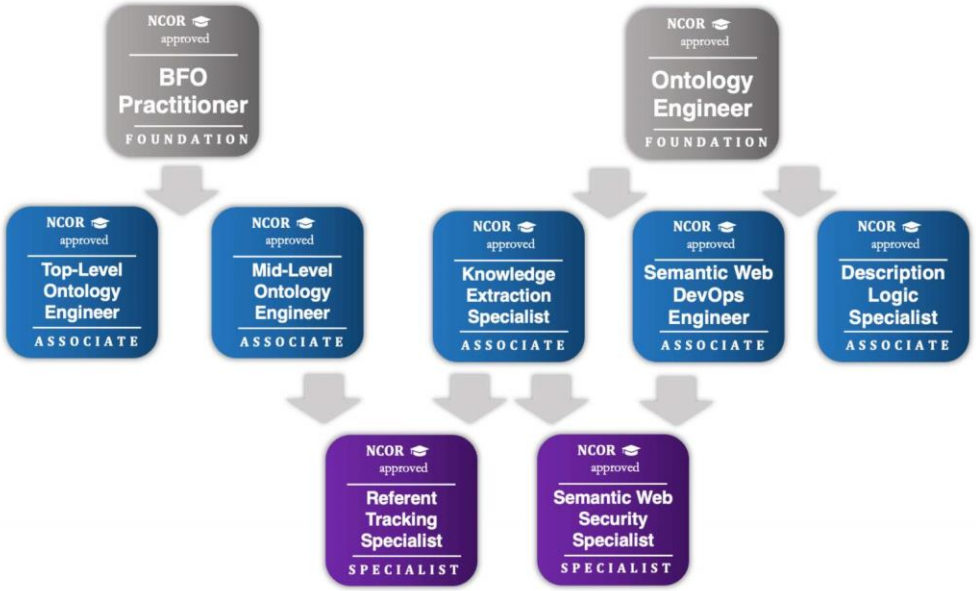
National Center for Ontological Research

NEED

As organizations attempt to leverage knowledge graphs in the interest of enterprise-level interoperability and data readiness goals, it is rather easy for them to reintroduce the problems they aim to solve. It is even easier for organizations to lack the experience needed to identify problematic knowledge graph development.

SOLUTION

NCOR Consulting offers knowledge graph architecture auditing, credentialing, training, and alignment services.



SERVICES

Quality Audits: Comprehensive evaluation reports of knowledge graph environments with actionable recommendations.

Credentialing: Certification badges for knowledge graphs that meet rigorous quality standards.

Training & Workshops: Train teams to integrate knowledge graphs into existing or novel workflows.

Alignment Consulting: Facilitate integration with industry standards or linked data environments.

WHY US?

Our team consists of knowledge representation pioneers and highly-trained experts with practical implementation experience across academia, government, and industry.

Our team is, moreover, highly regarded by organizations involved in this space, with an extensive track record of valuable assessments

Future of Education

national center for ontological research
ncor
academy

- Our **education and dissemination strategies** will continue to develop as we engage with a broader student base

- We will explore establishing **satellite schools** of applied ontology outside of UB

- As well as further solidify the role of NCOR as **the credentialing body** for the field of applied ontology

Outline

national center for ontological research

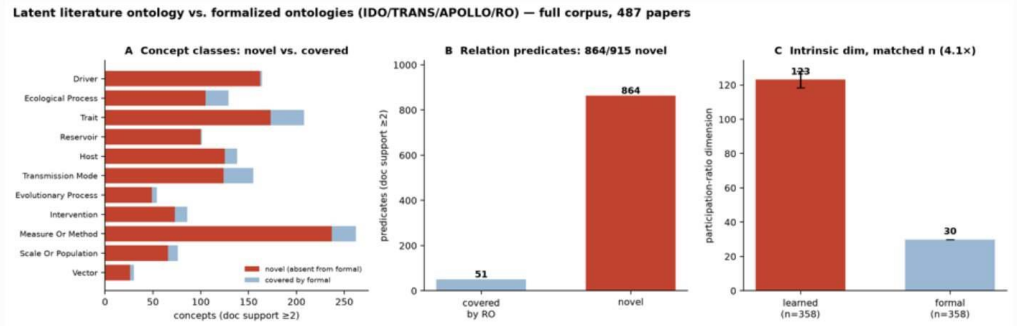
ncor academy

- Research and Development
- The Structure of Scientific Revolutions
- A New Era

Fox News guest Aaron Cohen used first lady Melania Trump's call for "pre-emptive intervention in identifying potential school shooters" on Wednesday to pitch America's "first-ever AI threat detection platform" for law enforcement.

"It scrapes the internet 24/7 using an Israeli-grade ontology to pull specific threat language and then routes it to local law enforcement," said Cohen, an American-Canadian actor and former Israel Defense Forces operative.

The project cost twenty-six dollars and change.



The latent ontology of spillover science is substantially richer than the formal ones. Learned de novo from the full text of 487 papers and compared to four formal ontologies, it contains 1,240 conceptual classes and 864 relation predicates wit… More

J.M. DRAKE

NEWSLETTER SIGN-UP

WSJ | CIO Journal

The Morning Download delivers daily insights and news on business technology from the CIO Journal team.

Preview Subscribe

Ghodsi said the "secret sauce" behind its new AI agents is a data context layer called Genie Ontology—essentially a graph of all knowledge in an organization, including data, content, apps, documents and people that is updated in real-time. Having that context leads to more accurate and faster responses from AI, as well as lower token costs, Ghodsi said.

"Agents can't simply be placed into the middle of a heterogeneous enterprise and expected to deliver scalable value," Dines says. "The organization first needs to build the substrate AI depends on: an ontology clearly defining the business entities, the actions required, and the cases, workflows, and processes involved, as well as make room for the exceptions that drive the day-to-day business. It's a cartography exercise, mapping the business so that AI can be anchored in the reality of the enterprise."

national
center for

Palantir

But why Palantir? You can't talk about the company's software programs without bringing up the company's proprietary ontology. As Karp said on the company's earnings call, "...ontology makes your enterprise accessible to AI." It's also what makes the bullish argument that it's hard to use conventional metrics to measure Palantir. Palantir would argue that among software-as-a-service (SaaS) companies, Palantir is not one of many; it's one of one.

Palantir Technologies Inc. (PLTR) Price Chart for Wednesday, February, 5, 2025



Proto-OKN Collaboration

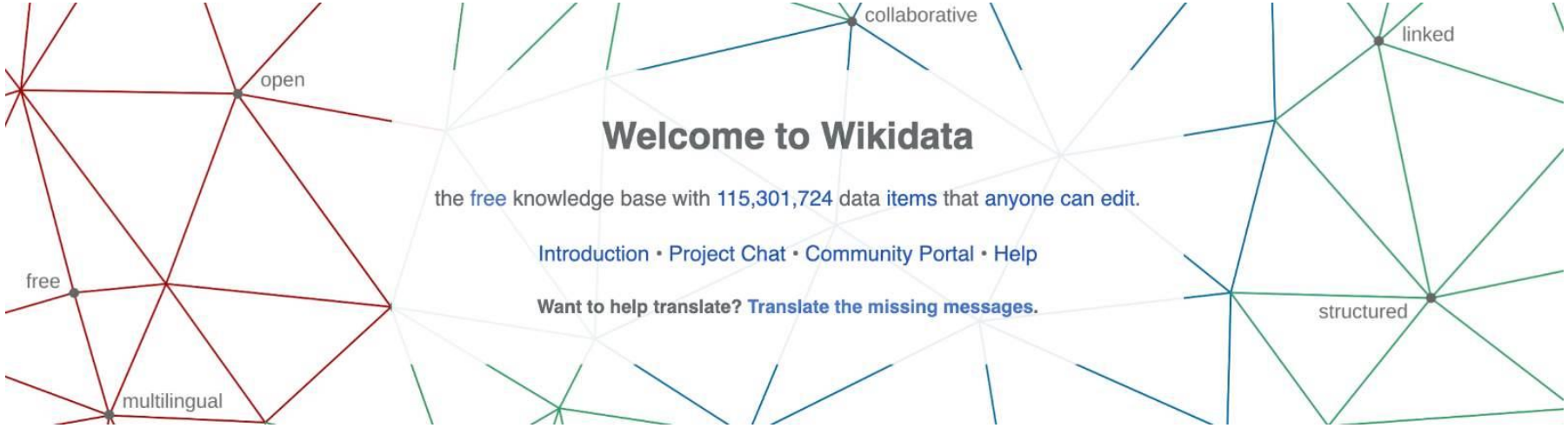
Facilitating the development and deployment of the first-ever Prototype Open Knowledge Network (Proto-OKN) to enable data-driven, artificial intelligence-based solutions for a broad set of societal and economic challenges

national center for ontological research

ncor academy

Decentralized Knowledge Graph (DKG)

OriginTrail Decentralized Knowledge Graph (DKG) presents a global, open data structure comprised of interlinked **Knowledge Assets structured in an RDF knowledge graph** hosted on an open, permissionless decentralized network of DKG nodes. It's designed to support the verifiable Internet for AI based on an open, permissionless knowledge economy in which knowledge is the primary asset class.



LLMs

- OpenAI's creation of ChatGPT has spurred the most extensive, most expensive test of the scalability hypothesis in history
- Shortly after dissemination, Nvidia rocketed in value, owing to recognition of the growing need for more compute power
- The Llama leak led to the proliferation of open-source models
- And subsequent advances in LLM research have resulted in rivals to OpenAI

Asset to Commodity

- In 2025, we inhabit a landscape in which LLMs are becoming **commodities** – a good that is interchangeable with goods of the same type, used as a foundation for production

- Rather than an **asset** – a resource that brings economic value for a specific organization

IN THE LAND OF COMMODIFIED LLMs...

IN THE LAND OF COMMODIFIED LLMs...
DATA IS KING



DeepSeek outperforms OpenAI's reasoning model at just 3% of the cost after President Trump's $500 billion Stargate AI initiative.

Key Features of DeepSeek

1. **Context-Aware Search:** Unlike traditional search engines, DeepSeek understands the meaning behind the words you type. It analyzes your query in context and delivers results that are more aligned with your needs. For example, if you're searching for "climate change," DeepSeek will not only provide articles but also pull up related data, graphs, expert opinions, and more nuanced content that speaks directly to your specific inquiry.
2. **Semantic Search:** At its core, DeepSeek utilizes **semantic search technology**, which focuses on the meaning of words in context rather than just the presence of keywords. This enables users to find results that are contextually relevant, even if the exact words you used aren't present in the source content.
3. **AI-Powered Knowledge Graph:** DeepSeek goes beyond a simple search engine. It uses an advanced knowledge graph to map relationships between entities, concepts, and topics. This graph makes it possible for DeepSeek to suggest further avenues of exploration, offering results that might not have been part of your original query but are highly relevant.
4. **Real-Time Data and Insights:** One of DeepSeek's standout features is its ability to offer real-time data and insights. Whether it's market trends, scientific discoveries, or breaking news, DeepSeek continuously updates its knowledge base to provide the most up-to-date information possible.
5. **Personalized Search Experience:** DeepSeek tailors its results to you, learning from your search history and preferences. This creates a more personalized and efficient search experience, as the platform becomes better at anticipating what you're looking for.

Key Features of DeepSeek

1. **Context-Aware Search:** Unlike traditional search engines, DeepSeek understands the meaning behind the words you type. It analyzes your query in context and delivers results that are more aligned with your needs. For example, if you're searching for "climate change," DeepSeek will not only provide articles but also pull up related data, graphs, expert opinions, and more nuanced content that speaks directly to your specific inquiry.
2. **Semantic Search:** At its core, DeepSeek utilizes **semantic search technology**, which focuses on the meaning of words in context rather than just the presence of keywords. This enables users to find results that are contextually relevant, even if the exact words you used aren't present in the source content.
3. **AI-Powered Knowledge Graph:** DeepSeek goes beyond a simple search engine. It uses an advanced knowledge graph to map relationships between entities, concepts, and topics. This graph makes it possible for DeepSeek to suggest further avenues of exploration, offering results that might not have been part of your original query but are highly relevant.
4. **Real-Time Data and Insights:** One of DeepSeek's standout features is its ability to offer real-time data and insights. Whether it's market trends, scientific discoveries, or breaking news, DeepSeek continuously updates its knowledge base to provide the most up-to-date information possible.
5. **Personalized Search Experience:** DeepSeek tailors its results to you, learning from your search history and preferences. This creates a more personalized and efficient search experience, as the platform becomes better at anticipating what you're looking for.

INTEROPERABILITY

INFORMATION QUALITY

KNOWN KNOWNS

"things that we're aware that we know"

KNOWN UNKNOWNS

"things that we're aware that we **don't** know"



UNKNOWN KNOWNS

"things that we're **unaware** that we know"

UNKNOWN UNKNOWNS

"things that we're **unaware** of and **don't** know"

INFORMATION QUALITY

Future of the Market

- Competitive advantage is becoming a matter of the best training data
- Open-source ontologies will be used to ground training and lower compute
- Competition is leading to a 'move fast and break stuff' mentality
- Creating opportunities for ontology **expertise and evaluation as a service**